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Assessing the Effectiveness of Large Language Model Support for Generating GDPR ROPA Documentation

Master Thesis at Ulm University

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Contents

1	Introduction	1
1.1	Kontext	1
1.2	Research Problem and Research Questions	2
1.3	3. Teil: Objectives und Contributions	3
2	Background	5
2.1	GDPR	5
2.2	Records of Processing Activities under Article 30 GDPR	6
2.3	Related Work	8
3	Methodology	20
3.1	User Study	20
4	Evaluation and Results	29
4.1	Usability — System Usability Scale	29
4.1.1	Per-Item Results	30
4.2	Cognitive Load — NASA Raw Task Load Index	32
4.2.1	Per-Subscale Results	34
4.3	Perceived AI Support — AI Questionnaire	37
4.3.1	Per-Item Means	37
4.3.2	Per-Item Descriptive Distributions	39
4.4	Mode Preference and Free-Text Feedback	41
4.4.1	Mode Preference Ranking	41
4.4.2	Free-Text Comments	42
4.5	Document Quality — Reference-Based NLP Metrics	44
4.5.1	Document-Level Results	44
	Lexical Metrics	44
	BERT-Based and Sentence-Embedding Metrics	48

Contents

4.5.2	Chapter-Level Results	52
4.5.3	Document-Level versus Chapter-Level Comparison	57
4.6	Document Quality — LLM-as-a-Judge	59
4.6.1	Per-Dimension Results	59
4.6.2	Cross-Dimension Patterns	61
4.6.3	Note on the Form Completeness Pattern	61
4.7	Summary of Findings	62
4.8	Participant Demographics	63
4.8.1	Numeric and Ordinal Variables	63
4.8.2	Categorical Variables	64
5	Discussion	67
5.1	User Study Discussion	67
5.2	Document Discussion	67
6	Conclusion	68
A	Study Scenario	69
A.1	Original (German)	69
A.2	English Translation	70
B	LLM-as-a-Judge Configuration	72
B.1	System and User Prompt	72
B.2	Rubric — Five Dimensions	74
B.3	Model Configuration and Limitations	74
C	Full Inferential Tables	77
C.1	Sample-Size Derivation	77
C.2	Survey Inferential Tables	77
C.3	NLP and Judge Inferential Tables	79
D	Inferential Statistics in Full	83
D.1	Survey Inferential Tests	83
D.1.1	Normality (Shapiro–Wilk, $n = 69$)	83
D.1.2	Friedman Omnibus ($n = 69$ matched triple)	84
D.1.3	Pairwise Wilcoxon Signed-Rank Post-hoc (Bonferroni $\alpha = 0.0167$)	84
D.1.4	Mode Preference Ranking	84

Contents

D.2	NLP Metric Inferential Tests	85
D.2.1	Kruskal–Wallis Sensitivity Check ($n = 113$, unbalanced) . . .	85
D.2.2	Friedman Omnibus ($n = 26$ matched triple)	86
D.2.3	Pairwise Wilcoxon Signed-Rank Post-hoc ($n = 26$, primary) .	86
D.2.4	Mann–Whitney U Sensitivity Check ($n = 113$, unbalanced) . .	86
D.3	Confidence–Quality Correlations	86
Bibliography		89

1 Introduction

1.1 Kontext

The General Data Protection Regulation (GDPR) requires organizations to maintain a Record of Processing Activities (ROPA), a formal document that outlines how personal data is collected, processed, stored, and shared. Creating and maintaining these ROPA documents is often a resource-intensive process, especially for small and medium-sized organizations that lack specialized legal or data-protection expertise. Previous research developed an application that leverages Large Language Models (LLMs) to support users in generating GDPR-compliant ROPA documentation. The application provides three distinct interaction modes, each offering different degrees of assistance from the LLM, from basic text suggestions to highly automated drafting of ROPA entries. This system aims to reduce the effort needed to create legally compliant documentation while maintaining transparency and user control. Although the application provides a functional framework for assisted ROPA generation, its effectiveness, usability, and impact on documentation quality have not yet been systematically evaluated. To ensure that such a system can be confidently integrated into practical compliance workflows, both qualitative and quantitative assessments are needed. These evaluations must determine whether users can efficiently and correctly create ROPA documents, how the varying levels of LLM support influence performance and trust, and whether the system produces outputs aligned with GDPR requirements.

1.2 Research Problem and Research Questions

The goal of this thesis is to conduct a comprehensive evaluation of the existing LLM-supported ROPA generation application. The work will focus on assessing how the different interaction modes affect user experience, efficiency, output quality, and perceived reliability.

RQ1:

- How do varying degrees of LLM support influence Novice users during ROPA document creation?
- How do novice users perceive the usability, efficiency, and satisfaction of the three interaction modes (Form, Ask, Chat)?
- How does cognitive load differ across the three modes for users with no prior GDPR knowledge?
- Which mode do novice users find most intuitive and prefer, and why?
- How confident do users feel about the quality of their generated documents in each mode? (This is now crucial - perception vs. reality)

RQ2:

- How do varying degrees of LLM support influence Document quality when used by novice users?
- How does document quality differ across the three generation modes based on semantic similarity to an expert-created reference (BERTScore)?
- Compare documents using an LLM
- Are all fields present? (Does my pipeline work)
- Is there a discrepancy between user confidence (RQ1) and document quality (RQ2)?

1.3 3. Teil: Objectives und Contributions

Thesis Organization

- **Chapter 1: Introduction**
 - Motivation, research questions, and contributions
- **Chapter 2: Background and Related Work**
 - GDPR and ROPA
 - Related work in AI-assisted legal document generation
- **Chapter 3: Methodology**
 - Brief overview of ROPAgen
 - User study design (within-subjects, balanced Latin Square)
 - Participant recruitment and demographic profile
 - Data collection instruments (System Usability Scale, NASA Raw Task Load Index, six-item Perceived AI Support questionnaire, mode-preference ranking, open-ended free-text comments)
 - Document quality assessment (reference-based NLP metrics, LLM-as-a-judge)
- **Chapter 4: Evaluation and Results**
 - Usability (SUS)
 - Cognitive Load (NASA R-TLX)
 - Perceived AI Support (six-item AI questionnaire)
 - Mode Preference and Free-Text Feedback (post-experiment ranking and open-ended comments)
 - Document Similarity (NLP Metrics, document- and chapter-level)
 - GDPR-Specific Quality (LLM-as-a-Judge)
 - Summary of Findings

- Participant Demographics
- **Chapter 5: Discussion**
 - Interpretation of findings against the research questions
 - Confidence–quality gap and implications
 - Limitations and comparison with related work
- **Chapter 6: Conclusion**
 - Summary of contributions and future work
- **Appendix**
 - Appendix A: Study Scenario (German original and English translation)
 - Appendix B: LLM-as-a-Judge Configuration (system/user prompt, rubric, model settings)

2 Background

2.1 GDPR

The General Data Protection Regulation (GDPR), implemented on May 25, 2018, represents the most stringent and comprehensive privacy and security framework in the global digital landscape [4]. While originated and enacted by the European Union, the regulation maintains an extraterritorial scope, imposing legal obligations on any organization—regardless of its physical location—that collects or processes data belonging to individuals within the EU. This landmark legislation is rooted in the 1950 European Convention on Human Rights, which established privacy as a fundamental right. It serves as a necessary evolution of the 1995 European Data Protection Directive, modernizing legal protections to address the rapid proliferation of cloud services, social media, and large-scale data breaches that have characterized the early 21st century. [4, 3]

Under the GDPR, the definitions of roles and data types are expanded to capture the complexity of modern informatics. Personal data is broadly defined as any information relating to an identifiable individual, encompassing everything from names and email addresses to biometric data, location information, and web cookies. The regulation distinguishes between the data subject (the individual), the data controller (the entity determining the "why" and "how" of processing), and the data processor (third parties handling data on behalf of a controller). To ensure ethical handling, all processing must adhere to seven core principles: lawfulness, fairness and transparency; purpose limitation; data minimization; accuracy; storage limitation; integrity and confidentiality; and accountability. This last principle is particularly vital, as it shifts the burden of proof to the organization, requiring documented evidence of compliance. [4]

The regulation mandates that processing is only legal if justified by one of six spe-

cific bases, most notably through consent, which must be freely given, specific, and unambiguous. Other justifications include contractual necessity, legal obligations, or a legitimate interest that does not override the subject's rights. Furthermore, the GDPR introduces the concept of data protection by design and by default, requiring that privacy considerations be integrated into the initial development phases of any product or service. Large-scale processors or public authorities are often required to appoint a Data Protection Officer (DPO) to oversee these internal processes and serve as a liaison with regulatory bodies. [4]

For individuals, the GDPR grants a robust suite of privacy rights, including the right to be informed, the right of access, the right to rectification, and the right to erasure (also known as the "right to be forgotten"). Subjects also possess the right to data portability and protections against automated decision-making and profiling. To enforce these standards, the regulation possesses significant "teeth"; authorities can levy harsh penalties for non-compliance. Consequently, the GDPR has transformed data privacy from a secondary IT concern into a primary strategic and legal priority for enterprises worldwide. [4].

2.2 Records of Processing Activities under Article 30 GDPR

Article 30 of the General Data Protection Regulation (GDPR) establishes the obligation for controllers and, where applicable, processors to maintain a Record of Processing Activities (RoPA). The RoPA functions as a core accountability instrument within the GDPR framework, operationalizing the principle that organizations must not only comply with data protection requirements but also be able to demonstrate such compliance to supervisory authorities. Its primary purpose is to provide a structured, transparent, and up-to-date overview of how personal data are processed within an organization, thereby enabling effective oversight, risk assessment, and regulatory control. [3, 12]

From a regulatory perspective, the RoPA serves several interrelated objectives. First, it enhances organizational awareness of data processing operations by requiring a systematic mapping of purposes, data flows, and safeguards. Second,

it supports data protection by design and by default, as organizations must explicitly consider data categories, retention periods, and security measures. Third, it enables supervisory authorities to efficiently assess compliance, particularly in the context of audits, investigations, or following personal data breaches. As such, the RoPA is not merely a formal record, but a living compliance document that reflects actual processing practices. [3]

Article 30 distinguishes between the obligations of data controllers and those of data processors. Controllers are required to document processing activities carried out under their responsibility, while processors must maintain records of processing performed on behalf of controllers. These records must be maintained in writing, including in electronic form, and must be made available to the competent supervisory authority upon request. Although the GDPR provides a limited exemption for organizations employing fewer than 250 persons, this exemption does not apply where processing is non-occasional, involves special categories of personal data, concerns criminal convictions, or is likely to result in risks to the rights and freedoms of data subjects. Consequently, in practice, many small and medium-sized organizations remain subject to the RoPA requirement. [3]

The content of a RoPA is explicitly defined by Article 30 GDPR and is intended to be concise, structured, and easily comprehensible. Excessive or irrelevant information should be avoided, as clarity and usability are essential for both internal governance and external supervision. At a minimum, a RoPA should include the following elements.

- Name and contact details of the data controller (and, where applicable, joint controllers, representatives, and the data protection officer).
- Purposes of the processing activities.
- Categories of data subjects.
- Categories of personal data processed.
- Categories of recipients, including recipients in third countries or international organizations.
- Details of transfers of personal data to third countries or international organizations, including safeguards where applicable.

- Time limits for the erasure of different categories of personal data.
- A general description of the technical and organizational security measures implemented pursuant to Article 32 GDPR.

Where an organization acts as a data processor, the RoPA must additionally reflect the processing carried out on behalf of each controller, including the identity of those controllers, the categories of processing involved, relevant international data transfers, and applicable security measures. In both cases, the RoPA should be sufficiently detailed to provide a faithful representation of processing activities, while remaining accessible to non-technical stakeholders. [3]

The GDPR does not prescribe a fixed update cycle for RoPAs; instead, it requires that records remain accurate and up to date. Consequently, any substantive change in processing purposes, data categories, recipients, or safeguards should trigger a review and update of the RoPA. Many organizations delegate the responsibility for maintaining the RoPA to a data protection officer, where one is appointed, to ensure consistency, avoid duplication of effort, and reduce the risk of omissions. [3, 9]

Finally, Article 30 emphasizes the external relevance of RoPA. Organizations must be prepared to present their records to supervisory authorities upon request, and such requests may be accompanied by demands for supplementary documentation, including privacy notices, consent records, or contractual arrangements. In this sense, RoPA operates as a foundational compliance artifact, anchoring broader data protection governance and demonstrating an organization's commitment to accountability under the GDPR. [3, 9].

2.3 Related Work

Open- vs. Closed-Source LLMs for ROPA Data Categories (von Schwerin et al., 2024)

The most directly relevant prior work is the systematic comparison by von Schwerin et al. [12], which targets the automation of GDPR compliance documentation, specifically the generation of data categories (e.g., contact information, payment

data) for Records of Processing Activities (ROPA). Although general legal AI research has covered tasks such as case retrieval and judgment prediction, the authors identify a significant research gap regarding the generation of structured, regulatory-compliant outputs, observing that existing legal benchmarks such as LawBench or LegalBench primarily evaluate static knowledge recall rather than the ability to produce documents that adhere to the precise formatting and terminology required by the GDPR.

Methodologically, the study employs a quantitative experimental design comparing four open-source models (Qwen2-7B, Meta-Llama-3-8B, zephyr-7b-beta, and orca_mini_v7_7b) against the closed-source GPT-4 as a baseline, structured around four experiments: basic performance, model-specific prompting, few-shot prompting, and Retrieval-Augmented Generation (RAG). To evaluate the documents the authors developed a comprehensive scoring metric called a Model Utility Score, weighting five criteria—Latency (weight 1, response speed), Structure and Style (weight 2, adherence to list formats and precise language), Grammar (weight 3, credibility and professional standards), Validity (weight 4, relevance of data categories to GDPR guidelines), and Contextual Understanding (weight 5, comprehensiveness and lack of redundancy in covering specific data fields). They also used two specialised benchmarks: a multiple-choice benchmark for contextual knowledge and a generation benchmark to compare LLM outputs against a reference database of correct data categories.

The findings demonstrate that open-source models, particularly Qwen2-7B, can achieve performance comparable to or even exceeding GPT-4 in generating structured ROPA data categories. Several nuances emerge from the synthesis: surprisingly, model-specific prompts aligned with pretraining formats did not necessarily improve performance, and simplicity and clarity in prompts often yielded better results; few-shot prompting had a limited impact on quality, while RAG showed mixed results, improving performance for categories explicitly found in the source text but causing models to struggle with novel categories and to ignore structural instructions such as list formatting. Overall, the study validates that open-source LLMs are cost-effective and privacy-preserving alternatives for ROPA documentation, since they allow for local deployment to satisfy strict GDPR data handling requirements.

The authors highlight several critical limitations for AI-driven compliance: LLMs still struggle with ambiguous legal language (e.g., terms like "necessary" or "adequate")

and variations in jurisdiction-specific terminology; the models rely on static datasets and may not reflect real-time updates to evolving GDPR regulations; a persistent risk of hallucination remains, where models generate plausible but legally incorrect information, necessitating a "human-in-the-loop" review process; and the evaluation was limited specifically to data category generation, which may not generalise to more complex GDPR tasks such as Privacy Impact Assessments.

Structured Indian Legal Document Generation (Nigam et al., 2025)

Nigam et al. [8] present *Structured Legal Document Generation in India: A Model-Agnostic Wrapper Approach with VidhikDastaavej*, addressing the automation of private legal document drafting within the Indian legal system. The authors identify a significant research gap: while previous work has extensively covered public legal tasks like judgment prediction and case summarisation, the generation of private, structured legal documents remains largely unexplored due to data confidentiality and the complexity of legal drafting standards.

The researchers developed NyayaShilp, a domain-adapted model based on LLaMA-2-7B which underwent continued pretraining (CPT) on Indian case laws and supervised fine-tuning (SFT) on a specialised dataset. Central to their approach is the Model-Agnostic Wrapper (MAW), a two-phase framework designed to improve long-form coherence: in Phase 1 (Section Title Generation) the model generates a structured list of section titles based on user input, which the user can manually refine; in Phase 2 (Section Content Generation) the system iteratively generates content for each section, using a vector database (ChromaDB) to store and retrieve summaries of previous sections to maintain logical flow and reduce hallucinations. The system is integrated into an interactive Human-in-the-Loop (HITL) interface that allows users to specify document types, customise sections, and refine AI-generated drafts. To support the work, the authors introduced VidhikDastaavej, a novel anonymised dataset of 489 private legal documents (e.g., petitions, affidavits, lease deeds). Their evaluation methodology mirrors a dual-evaluation structure: quantitative lexical and semantic metrics including ROUGE, BLEU, METEOR and BERTScore for semantic similarity between generated and reference texts; an automatic LLM-based evaluation using G-Eval, a GPT-4-based framework, to assess

coherence, factual accuracy, and completeness; and a qualitative expert assessment in which three legal experts evaluated documents on a 1–10 Likert scale for factual accuracy and completeness, with inter-annotator agreement (IAA) measured via Intraclass Correlation and Krippendorff’s Alpha.

The findings indicate that direct fine-tuning (SFT) on small datasets can actually lead to performance degradation and hallucinations. In contrast, the structured wrapper approach (MAW) significantly enhanced document quality, achieving scores comparable to GPT-4o even when using smaller open-source models, and experts confirmed that the wrapper-assisted models produced more coherent, well-structured, and legally valid documents compared to base models.

The authors note several limitations: the training data was imbalanced, with a heavy focus on case judgments, leading to difficulties in generating underrepresented legal formats like contracts or agreements; while the wrapper reduced inconsistencies, it did not entirely eliminate hallucinations, suggesting a need for more robust fact-verification modules; the study primarily relied on expert evaluation, with large-scale user feedback from practising lawyers and broader usability testing across different jurisdictions identified as necessary future steps; and computational constraints meant the research did not fully explore larger architectures (e.g., LLaMA-3-70B) for fine-tuning.

LLM-Assisted Extraction of GDPR Requirements (Abualhaija et al., 2025)

Abualhaija et al. [1], in *LLM-assisted Extraction of Regulatory Requirements: A Case Study on the GDPR*, address the knowledge gap between legal interpretation and software development, focusing on the extraction of technical requirements from the GDPR. They argue that while the GDPR establishes fundamental data subject rights, it often lacks the detailed technical guidance necessary for software implementation, leading to low compliance rates. The authors identify a specific research gap in the requirements engineering (RE) literature: most studies provide only generic mentions of the GDPR, with very few investigating specific data subject rights—such as the right of access (ACC) and the right to portability (PRT)—in depth. The study further explores the potential for LLMs to bridge this gap by

leveraging additional legal sources (guidelines, case law, etc.) beyond the primary GDPR text.

The authors used a mixed-methods approach combining qualitative manual extraction and quantitative automated generation. As a manual baseline, in collaboration with six legal and RE experts, they extracted 108 reference requirements (61 for ACC; 47 for PRT) from mandatory legal sources, advisory guidelines, and academic literature. Their automated approach, XTRAREG, uses GPT-3.5 and GPT-4o combined with Retrieval-Augmented Generation (RAG), and tests different prompting techniques including Zero-Shot Learning (ZSL) and Chain-of-Thought (CoT) to instruct the LLM to generate requirements informed by indexed legal documents. The dual evaluation framework comprises a quantitative side—automated NLP metrics comparing generated text to the manual ground truth, including precision/recall-oriented metrics (BLEU, ROUGE-L, METEOR) and the semantic similarity metric BERTScore—and a qualitative side, in which legal experts performed an extrinsic assessment of the generated requirements based on Correctness (factually accurate), Groundedness (valid legal references), Plausibility (reasonable rationale), and Coverage (percentage of reference requirements identified).

GPT-4o significantly outperformed GPT-3.5, particularly in the volume and completeness of generated requirements. XTRAREG achieved a correctness accuracy of 81.8% for ACC and 85.7% for PRT; the inclusion of RAG significantly improved performance across all metrics compared to "ablation" runs in which the LLM relied only on its pre-trained knowledge; the models achieved high BERTScores, indicating that they successfully captured the semantic meaning of the requirements even when they did not match the manual baseline verbatim (as shown by lower BLEU scores); and despite being exposed to various sources via RAG, the LLMs showed a strong tendency to generate requirements predominantly from the GDPR itself, often overlooking more specific technical details found in supplementary guidelines.

Several limitations of using LLMs for regulatory tasks are highlighted: while the requirements generated were often correct, the LLM only covered a fraction of the total requirements identified by human experts; the LLM frequently adopted an incorrect perspective, writing from the viewpoint of the data subject (e.g., "the user may...") rather than the system (e.g., "the system shall..."); some outputs were non-actionable, providing verbatim legal excerpts rather than technical requirements, or failing to account for legal exceptions; and the models occasionally

suggested incomplete or misgrounded references, sometimes citing recitals (which are non-binding in civil law systems) instead of specific normative articles.

Fine-Tuned LLM for Chinese Legal Drafting (Lin and Cheng, 2024)

Lin and Cheng [7], in *Legal Documents Drafting with Fine-Tuned Pre-Trained Large Language Model*, focus on the automated drafting of legal documents, specifically targeting "criminal facts" descriptions in Traditional Chinese fraud case verdicts. The authors identify a significant research gap in the legal domain: traditional Natural Language Processing (NLP) methods rely heavily on manually annotated datasets, which are difficult and costly to obtain for specialised legal tasks; furthermore, most existing LLMs are trained on general text, leading to suboptimal performance when generating documents that require specific legal terminology and rigid formatting. A critical secondary gap is the privacy and security risk associated with third-party AI systems, as legal professionals are often reluctant to use cloud-based LLMs because they handle sensitive information.

The researchers employed a quantitative approach to model training and a structural qualitative-to-quantitative metric for document evaluation. To address privacy concerns, the authors fine-tuned an open-source BLOOM-560M model on a local machine (RTX 3090). They used a large unlabelled dataset of 74,823 fraud verdicts from the Taiwan Judicial Yuan (2011–2021) without requiring Chinese word segmentation, lowering the threshold for processing Chinese legal text. A core methodological contribution is a structural-correctness evaluation based on legal constitutive elements: the authors defined a specific order for document elements—Subject of Crime (LEO_SOC), Subjective Legal Element (LEO_SLE), Act (LEO_ACT), Victim (LEO_VIC), Causation (LEO_CAU), and Result of Hazard (LEO_ROH)—to evaluate whether the LLM-generated draft follows the correct legal logic and format. The model was further evaluated using perplexity and training/validation loss.

The study demonstrates that local fine-tuning of a small-scale LLM (560M parameters) significantly improves its ability to generate legal drafts compared to the base model, which failed to produce relevant legal content. The fine-tuned model successfully generated drafts that adhered to the pre-defined structural order of legal

elements (Equation 1), confirming the feasibility of using LLMs for specialised legal drafting. Synthesising these results, the paper shows that domain-specific performance can be achieved without massive computational resources or cloud-based infrastructure, making it a viable solution for privacy-sensitive legal applications.

The authors acknowledge several limitations: the experimental dataset is relatively small and focused only on fraud cases, requiring broader validation; perplexity alone may not fully reflect the practical utility or quality of the documents in a real-world application; fine-tuning larger models (e.g., 1.1B parameters) requires substantial GPU memory beyond standard consumer-grade hardware; and the model occasionally produces irrelevant or non-existent terms (e.g., the word "nueve"), a persistent challenge in natural language generation.

AI-Powered Legal Documentation Assistant (Vayadande et al., 2024)

Vayadande et al. [15] describe an *AI-Powered Legal Documentation Assistant* focused on the automation of legal document processing, specifically addressing document generation, comprehension, and abnormality detection. The authors identify a significant research gap in the inherent complexity of legal systems, which traditionally rely on manual, time-consuming processes prone to human error, and characterise existing document processing systems as inadequate because they cannot keep pace with the precision-oriented nature of legal documentation or the intricacies of specialised legal language.

The authors developed a system using ChatGPT 3.5 integrated with OpenAI Embeddings and the LangChain framework. The architecture follows a multi-stage pipeline: data assembly of multilingual legal styles, data cleansing, and the use of FAISS vector stores for similarity searches. It employs PyPDF and Amazon Textract for document parsing, and Named Entity Recognition (NER) combined with sentiment analysis to identify inconsistencies. Evaluation is primarily quantitative, featuring a comparative analysis of the proposed LDA against existing market solutions like Kira Systems and Evisort, with metrics including accuracy percentages, time efficiency (hours), and document processing speed (pages per minute). The study additionally incorporates a qualitative element through a debugging phase

that uses feedback and "balanced comments" from legal experts to refine the system's practical utility.

The study reports that the AI-powered assistant significantly outperforms traditional methods: an accuracy rate of 93–97 %, comparable to or higher than established legal AI tools; a reduction in the time required for document tasks from 20 hours (manual) to 5 hours (AI-assisted); and an error rate lowered to 5 %, compared to a 15 % error rate in manual processing. The synthesis of these results suggests that combining LLMs with specialised NLP toolkits (such as NER and sentiment analysis) provides a "force of change" for making legal practices more frictionless.

Despite the high performance, the authors note several limitations: the system still lacks "absolute accuracy" and may struggle with the extreme nuances of certain judicial texts; while designed for multilingual potential, it does not yet apply to a wide enough variety of languages, and accuracy is highly dependent on the quality of input, such as the clarity of audio recordings or the success of digitising handwritten notes; and using different text extraction tools for various formats (PDF vs. DOC) can lead to inconsistencies in how information is expressed.

AI-Powered Interactive Legal Chatbot (Surya et al., 2025)

Surya et al. [14] describe an *AI-Powered Interactive Legal Chatbot for the Department of Justice*, focusing on the Indian public legal assistance domain and specifically targeting the significant gap in accessibility to understandable and affordable legal information. They identify a research gap in providing scalable, real-time, and multilingual support for citizens who struggle with complex legal jargon and the financial burden of professional counsel, and aim to democratise legal information, particularly for marginalised communities and those with limited resources.

The methodology centres on the development of a three-tier conceptual and technical framework for an AI-powered chatbot. The system architecture comprises a frontend (a JavaScript-powered single-page application for user interaction and document uploads), a backend (a Python Flask orchestrator handling API requests and language translation), and an LLM service (LLaMA 3.1 accessed via the Ollama framework, using specific prompt engineering for legal queries and document summarisation). For evaluation, the paper uses qualitative illustrative use cases to

demonstrate functionality rather than a rigorous quantitative user study. These use cases simulate real-world scenarios, such as a user inquiring about consumer complaint procedures or another requesting a summary of a First Information Report (FIR), and also showcase multilingual interaction by processing a query in Hindi, translating it to English for the LLM, and returning the response in Hindi.

The primary result is the successful integration of real-time query processing and document summarisation within a unified legal assistant. The authors synthesise these features to show how such a tool can provide 24/7 availability, reduce cost barriers, and simplify complex legal language for laypersons. A significant contribution is the multilingual capability, which allows the system to reach a broader demographic by breaking down language barriers, and the system also emphasises transparency through citation support, providing users with relevant legal references and government portal links to verify information.

The paper identifies several critical limitations: LLMs may generate plausible but incorrect information, which is particularly risky in the legal domain where precision is paramount; the system may struggle with the subtleties of legal language, potentially leading to oversimplifications or misinterpretations in summaries of complex documents like FIRs or petitions; while the system handles sensitive personal data, current implementations require more robust measures like encryption and access controls to comply with India’s Digital Personal Data Protection Act; and there is an inherent risk of AI bias and a fine line between providing information and offering actionable legal advice, which users might misinterpret despite disclaimers.

JuDGE: Judgment Document Generation Benchmark (Su et al., 2025)

Su et al. [13], in *JuDGE: Benchmarking Judgment Document Generation for Chinese Legal System*, focus on the automated generation of complete legal judgment documents within the Chinese legal system, a task that demands complex legal reasoning and the integration of professional knowledge. While previous work in Legal Judgment Prediction (LJP) primarily formulated the problem as a classification task—predicting discrete labels for charges or sentences—this paper addresses the generative gap by requiring models to produce structurally coherent and legally

sound full-text documents. The authors identify a critical research gap: the absence of standardised benchmarks and automated evaluation frameworks, which previously necessitated time-consuming and unscalable expert annotations.

The authors introduce JuDGE (Judgment Document Generation Evaluation), a benchmark designed to mirror the judicial syllogism process: using statutes and precedents as the major premise and case facts as the minor premise to reach a legal conclusion. The benchmark includes 2,505 fact-judgment pairs from real criminal cases, supplemented by an external corpus of over 55,000 statutory articles and 103,000 past judgments. The study employs a multi-dimensional automated framework to assess quality across four perspectives: *Penalty Accuracy*, measuring the precision of prison terms and fines using a normalised absolute difference score; *Convicting Accuracy*, using Recall, Precision, and F1 scores to evaluate charge identification; *Referencing Accuracy*, assessing the correctness of statutory citations; and *Documenting Similarity*, evaluating semantic alignment with ground truth via METEOR and BERTScore to capture nuanced legal reasoning beyond simple lexical matching.

The experiments compared several baselines, including direct generation, few-shot in-context learning (ICL), supervised fine-tuning (SFT), and Multi-source Retrieval-Augmented Generation (MRAG). General and legal-domain LLMs performed poorly when used directly or via ICL, often failing to incorporate case-specific facts; however, SFT significantly improved performance across all models, making previously "unusable" legal LLMs competitive. The proposed MRAG approach, which retrieves both relevant statutes and similar cases, achieved the highest performance in convicting accuracy. Synthesising the results, the authors observe that while advanced techniques like RAG and SFT enhance document quality, current LLMs still struggle to consistently produce documents that match the legal rigor of professional ground truths, indicating that the task remains highly challenging.

The paper notes several limitations. First, while MRAG improves recall for legal references, it often suffers from lower precision because current LLMs struggle to filter out irrelevant statutes provided by the retriever. Second, the automated metrics provided are currently focused on criminal law, which may limit their immediate applicability to other legal domains. Finally, the authors acknowledge that metrics like METEOR and BERTScore are proxies for semantic consistency and may not fully capture the absolute legal validity that a human expert would provide.

CaseGen: Multi-Stage Legal Case Documents (Li et al., 2025)

Li et al. [5], in *CaseGen: A Benchmark for Multi-Stage Legal Case Documents Generation*, focus on the Chinese legal domain, specifically the automation of drafting official judicial records, which traditionally requires significant manual effort and domain expertise. They identify a critical research gap: while LLMs show promise in legal applications, existing benchmarks are limited to general text tasks (like summarisation) or simple discriminative legal tasks (like judgment prediction or case retrieval), and there was previously no benchmark capable of capturing the complexity, professional rigour, and multi-stage nature of drafting real-world legal documents.

CaseGen uses a dataset of 500 real-world legal cases annotated by five legal experts to ensure authenticity and completeness. The methodology is defined by two core components. The first is a multi-stage task design: instead of generating a document in a single step, the benchmark breaks the process into four distinct tasks—drafting defence statements, writing trial facts, composing legal reasoning, and generating judgment results—an approach that aligns with the actual logic of legal writing and allows for a more nuanced assessment of an LLM’s strengths and weaknesses. The second is a hybrid evaluation framework. On the quantitative side, traditional metrics like ROUGE-L and BERTScore are used, but noted for their inability to capture logical coherence or factuality. The primary evaluation is an LLM-as-a-Judge pointwise scoring system (1–10) where an LLM (GPT-4o) acts as a judge, using Chain-of-Thought (CoT) reasoning, task-oriented criteria (accuracy, completeness, logic), and reference-based comparisons to ground truth documents. The effectiveness of the LLM judge was validated through human expert annotations, showing a 75 % Spearman correlation, which suggests that LLM-as-a-judge is a reliable alternative for large-scale legal text evaluation.

Surprisingly, general-domain LLMs (such as GPT-4o and Qwen2.5) consistently outperformed legal-specific models like ChatLaw or LexiLaw; the authors suggest that specialised models may suffer from limited base-model reasoning or information loss due to shorter context windows. Despite the capabilities of modern LLMs, most models achieved unsatisfactory scores (below 6 out of 10), indicating that existing AI still struggles with the logical rigor and complex reasoning required for professional legal drafting. The study found that the LLM-as-a-judge score had sig-

nificantly higher consistency with human experts than lexical matching metrics like ROUGE-L.

The authors note several limitations: the benchmark is currently limited to the Chinese legal system, meaning the findings may not generalise to different legal frameworks or document formats; while highly correlated with humans, LLM judges can still be influenced by inherent biases or adversarial inputs and cannot yet fully replace nuanced professional judgment; and the study emphasises that AI should remain a supportive tool for efficiency rather than a substitute for human legal expertise, especially given the risks of hallucinations in high-stakes environments.

Synthesis

Across the eight works reviewed, two patterns emerge that frame the present thesis. First, dual evaluation—combining surface-overlap NLP metrics (BLEU, ROUGE, METEOR), semantic-similarity metrics (BERTScore), and either expert assessment or LLM-as-a-Judge scoring—is becoming the *de facto* standard for legal-document generation, and is the methodological template adopted here. Second, while several works explore architectures (RAG, fine-tuning, structured wrappers) that improve generated-document quality on benchmarks, none of them measure how *novice users* interact with such systems on a GDPR-specific task or how user confidence relates to actual document quality. The thesis fills that gap by combining a within-subjects user study with a dual-evaluation pipeline against a single shared scenario.

3 Methodology

3.1 User Study

Study Design

The study followed a within-subjects experimental design with counterbalanced ordering, using a Balanced Latin Square. The recruited sample consisted of $N = 73$ university students, all novice users with no prior GDPR experience, and each session lasted approximately 90 minutes per participant. Counterbalancing was implemented through a Balanced Latin Square with six groups, reflecting the three AI modes tested in randomised sequences; participants were randomly assigned to one of the six groups, and each completed all three modes in a predetermined counterbalanced order. This design controls for order effects, learning effects, and fatigue.

Task Scenario

A single standardised scenario was used across all participants and modes to ensure consistency. The scenario described a relatable university-based data processing activity covering all GDPR Article 30 fields, ensuring comparability across conditions while remaining accessible to participants without legal expertise. The underlying LLM was Mistral Large 3 (model identifier `mistralai/mistral-large-2512`), held constant across all modes and routed via OpenRouter.

Participants

Recruitment yielded $N = 73$ university students. Participants were screened for minimal to no GDPR knowledge (expected and required), prior AI tool usage experience (documented), general technology comfort level (assessed), and the absence of any professional legal or compliance background. As part of onboarding, participants received a brief standardised introduction to GDPR and ROPA concepts before beginning the tasks, and were shown an example of a high-quality ROPA document as reference material prior to starting.

Data Availability and Sample Sizes

Due to technical data loss and incomplete submissions during data collection, the effective sample sizes differ across analysis streams. The canonical sample sizes used throughout this thesis are summarised below; all counts are derived from the raw data files committed to the repository and documented in full in `python/output/sample_size`

- **Survey, recruited $N = 73$ / valid $N = 70$** — 73 participants completed the study; 70 retained after dropping cases with invalid consent or quality flags. Per-mode sample size: **$n = 69$** for each of Form, Ask, and Chat (one participant has a valid Form response but no Ask/Chat). The matched-triple survey subset (participants with valid responses for all three modes) is **$n = 69$** ; this is used for within-subjects Friedman and Wilcoxon signed-rank tests on SUS, R-TLX, confidence items, and the perceived AI support composite.
- **Documents, 113 generated ROPAs** — distributed across modes as 37 (Form), 37 (Ask), and 39 (Chat); used for all NLP metric and LLM-as-a-judge analyses at the document level. The 113 documents represent **47 unique participants**; **28 participants** have at least one document in each of the three modes (loose match), and **26 participants** have exactly one document per mode (strict matched triple). The strict **$n = 26$** set is used for all within-subjects tests on document-quality measures (Friedman, Wilcoxon signed-rank); the unbalanced full sample ($n = 113$) is used only for between-groups sensitivity checks (Kruskal–Wallis, Mann–Whitney U).
- **Survey $\times$ NLP join, 106 rows** — inner join on participant and mode yields

106 rows (Form 34, Ask 34, Chat 38) across **44 unique participants**. Per-mode joined samples ($n = 34 / 34 / 38$) are used for per-mode Spearman correlations between confidence and quality, which do not require complete triples.

- **Triangulated sample, $n = 27$** — participants with survey $\times$ NLP coverage in all three modes. This is the canonical N for confidence–quality correlation analyses that require within-subjects matching across all three modes simultaneously.

The reconciliation between these canonical N s and earlier reporting is as follows: 47 corresponds to unique participants with at least one judge-evaluated document, 28 to participants with at least one document per mode (loose match), and 44 to unique participants in the survey $\times$ NLP join. The canonical figures used in the inferential analyses are **$n = 26$** for paired NLP/judge tests and **$n = 27$** for triangulated correlations across all three modes.

Descriptive statistics (means, medians, standard deviations) are computed on the full available N per mode for each analysis stream. Within-subjects inferential tests are restricted to the matched-triple subsets specified above. Where the unbalanced full sample makes matching impossible, between-groups sensitivity checks are reported alongside the matched analyses (see *Statistical Analysis*, below).

Data Collection

Quantitative measures were collected after each mode. The System Usability Scale (SUS) was administered as ten standardised questions on a 5-point Likert scale, and the Raw Task Load Index (R-TLX) captured six cognitive load dimensions, also on a 5-point Likert scale. In addition, participants completed a six-item Perceived AI Support instrument on a 5-point Likert scale:

1. *Perceived Usefulness*: “The LLM was helpful in creating this document.”
2. *Efficiency*: “This mode let me complete the ROPA faster than I could on my own.”
3. *Transparency*: “I understood why the AI suggested certain GDPR fields.”

4. *Reliance Comfort*: “I felt comfortable relying on the AI’s suggested categories.”
5. *Task-Specific Confidence (primary confidence measure)*: “I am confident the AI identified the appropriate legal basis.”
6. *Intention to Use*: “I would use this mode again for similar work.”

Item 5 is treated as the **primary confidence measure** throughout the analysis: it is the only item that asks participants to judge the substantive correctness of the AI’s output (the legal basis under Article 6 GDPR), and is therefore directly comparable to the document-quality signals from the NLP and judge pipelines. The remaining five items (usefulness, efficiency, transparency, reliance comfort, intention to use) capture distinct facets of *perceived AI support* rather than confidence in output correctness, and are reported as a secondary composite. Free-text qualitative feedback was also collected after each mode, and total session duration was recorded as a single value per participant; per-mode timing was not recorded. After completing all three conditions, participants ranked all three modes by overall preference in a post-experiment ranking task.

Quality Assessment

NLP Metrics

All participant-generated documents (73 participants $\times$ 3 modes = 219 documents) are compared against a single reference document using a suite of automated NLP metrics. The reference document was created by the study author using the Form mode of ropagen, applying the same standardised scenario presented to participants. This document is treated as the authoritative baseline — a root-of-trust representing a careful, complete application of the most structured interaction mode. It should be noted that this introduces an inherent framing effect: NLP metrics measure similarity to Form-style output, which may disadvantage the more conversational outputs of Ask and Chat modes independently of their substantive quality.

- **BLEU** (Bilingual Evaluation Understudy): n-gram precision with brevity penalty
- **ROUGE-1, ROUGE-2, ROUGE-L**: unigram overlap, bigram overlap, and longest common subsequence F1

- **METEOR**: precision–recall balance with synonym matching
- **BERTScore** (Precision, Recall, F1): token-level semantic similarity using ModernBERT embeddings
- **SBERT Cosine Similarity**: document-level semantic similarity using sentence embeddings

Metrics are computed at both the document level (aggregated across all Article 30 sections) and the chapter level (per individual GDPR section). Results are aggregated by mode to establish comparative quality baselines.

LLM-as-a-Judge Evaluation

Documents are additionally evaluated using an LLM-as-a-judge approach, in which **Gemini 3.1 Pro Preview** (model identifier `google/gemini-3.1-pro-preview`, accessed via OpenRouter at temperature 0) serves as an automated evaluator assessing GDPR-specific quality. The judge is positioned as a *supplementary* evaluator: it is a single-pass, single-model assessment without inter-rater replication or legal-expert validation, and its scores are reported alongside, not in place of, the NLP metrics. The reference-based NLP metrics remain the primary RQ2 evidence; the judge complements them by providing a reference-free signal that captures GDPR-specific quality properties (legal correctness, hallucination, scenario faithfulness) which similarity-based metrics cannot directly observe. The judge evaluates each document across five dimensions on a 1–5 scale:

1. **Completeness**: Coverage of all required Article 30(1) fields (controller identity, DPO contact, purposes, legal basis, data categories, recipients, retention periods, technical and organisational measures)
2. **Scenario Faithfulness**: Adherence to the specific details of the study scenario rather than generic or hallucinated content
3. **Legal Correctness**: Appropriate use of GDPR terminology and accurate citation of relevant articles
4. **Hallucination (inverse)**: Absence of invented facts not present in the scenario (scored inversely: 5 = no hallucinations)

5. **Overall:** Holistic assessment of the document’s usability as a real-world ROPA record

The judge evaluation is applied to all 113 retained documents. The full evaluation rubric and system prompt used for the judge are provided in Appendix B.1. The interpretive narrative summarising judge scores in the accompanying analysis artefact (`judge_interpretation.md`) was drafted by Claude Sonnet 4.6 from the raw per-document scores and rationales as a working aid; the interpretation reported in this thesis is authored by the researcher directly from the same raw judge scores.

Confidence–Quality Gap Analysis

To investigate whether perceived confidence in the AI-generated output corresponds to actual document quality, self-reported confidence is correlated with both NLP metric scores and LLM-as-a-judge scores. The **primary confidence measure** is the single Likert item on legal-basis confidence (item 5 of the perceived AI support instrument: “I am confident the AI identified the appropriate legal basis”), chosen because it is the only survey item that asks participants to judge the substantive correctness of the AI’s output. The 6-item Perceived AI Support composite is reported as a **secondary** analysis to capture broader attitudinal facets (usefulness, efficiency, transparency, reliance comfort, intention to use). Both measures are correlated against three signals: confidence against NLP metrics (reference-based lexical and semantic similarity); confidence against LLM judge scores (GDPR-specific holistic quality); and NLP metrics against LLM judge scores (agreement between the two automated assessment systems). Analyses are restricted to participants for whom data from all three sources (survey, NLP pipeline, judge evaluation) are available; the exact triangulated N is reported in the *Data Availability and Sample Sizes* subsection.

Statistical Analysis

A significance threshold of $\alpha = 0.05$ is applied to all tests. All within-subjects comparisons across the three modes follow a two-step procedure: a Friedman test as the omnibus test, followed by pairwise Wilcoxon signed-rank tests with Bonferroni

correction for post-hoc analysis. Where the unbalanced full sample makes within-subjects matching impossible (e.g. when only one or two documents are available per participant), Mann–Whitney U tests are reported as a between-groups sensitivity check on the corresponding mode pair. Effect sizes are reported as ε^2 for the Friedman / Kruskal–Wallis omnibus tests, Kendall’s W as the coefficient of concordance for Friedman, and rank-biserial correlation r for pairwise Wilcoxon signed-rank tests.

For RQ1 (User Experience and Preference), the Friedman test is applied to SUS scores, the R-TLX composite, and the confidence question composites across modes; post-hoc pairwise Wilcoxon signed-rank tests with Bonferroni correction identify which mode pairs differ significantly; and mode preference distributions are reported as frequencies and rankings. For RQ2 (Document Quality and Confidence Calibration), the Friedman test is applied to NLP metric scores and LLM-as-a-judge scores across modes, with Bonferroni-corrected pairwise Wilcoxon signed-rank tests as post-hoc analysis where the omnibus is significant. Spearman rank correlation (ρ) is used for the confidence–quality gap analysis (non-parametric; appropriate for ordinal-to-continuous relationships).

Definitions of Statistical Tests and Effect Sizes

The following definitions apply to every measurement reported in the Evaluation chapter. They are listed once here rather than re-introduced for every instrument.

Shapiro–Wilk normality test. Tests the null hypothesis that a sample is drawn from a normal distribution. The statistic is

$$W = \frac{\left(\sum_{i=1}^n a_i x_{(i)}\right)^2}{\sum_{i=1}^n (x_i - \bar{x})^2}, \quad (3.1)$$

where $x_{(i)}$ are the order statistics and a_i are tabulated constants. W close to 1 indicates normality; small p -values reject normality. In this study, normality is rejected for nearly every survey measure, motivating the non-parametric tests below.

Friedman test. Non-parametric repeated-measures omnibus test for $k \geq 3$ related samples. Each participant's k values are ranked within-subject; the test statistic on n participants across k conditions is

$$\chi_F^2 = \frac{12}{nk(k+1)} \sum_{j=1}^k R_j^2 - 3n(k+1), \quad (3.2)$$

where R_j is the sum of ranks for condition j . Under H_0 (no difference among conditions), χ_F^2 is asymptotically χ^2 distributed with $k - 1$ degrees of freedom. A significant Friedman statistic motivates pairwise post-hoc comparisons.

Wilcoxon signed-rank test (post-hoc). Non-parametric test for two related samples, applied as a pairwise post-hoc to a significant Friedman omnibus. Differences $d_i = x_i - y_i$ are ranked by absolute value; the statistic

$$W = \min(W_+, W_-), \quad W_+ = \sum_{d_i > 0} \text{rank}(|d_i|), \quad W_- = \sum_{d_i < 0} \text{rank}(|d_i|). \quad (3.3)$$

With three modes there are three pairwise comparisons; the Bonferroni-corrected significance threshold is therefore $\alpha' = 0.05/3 = 0.0167$.

Mann–Whitney U test (between-groups sensitivity check). Non-parametric two-sample test on independent groups, used here as a sensitivity check on the unbalanced full sample where matched-pair tests are not applicable. With n_1 and n_2 the two group sizes and R_1 the sum of ranks in group 1,

$$U = R_1 - \frac{n_1(n_1 + 1)}{2}. \quad (3.4)$$

Kruskal–Wallis H test. Non-parametric one-way ANOVA analogue for k independent groups, used in the Evaluation chapter as a between-groups sensitivity check on the unbalanced full document sample. With N total observations, n_j observations in group j , and R_j the sum of ranks in group j ,

$$H = \frac{12}{N(N+1)} \sum_{j=1}^k \frac{R_j^2}{n_j} - 3(N+1). \quad (3.5)$$

Chi-square goodness-of-fit test. Used here on the distribution of rank-1 votes in the mode-preference data, against a uniform null. With O_i observed and E_i expected counts in k categories,

$$\chi^2 = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i}. \quad (3.6)$$

Effect sizes. Effect sizes complement p -values by quantifying the magnitude of the difference, independent of sample size.

- Kendall's W (coefficient of concordance) is the Friedman test's effect size on $[0, 1]$: $W = \chi_F^2 / (n(k - 1))$. Conventional bands: 0–0.1 weak, 0.1–0.3 moderate, 0.3–0.5 strong.
- Rank-biserial correlation r is the matched-pairs effect size for the Wilcoxon signed-rank test, given by $r = (W_+ - W_-) / (W_+ + W_-)$. Conventional bands: $|r| < 0.1$ negligible, 0.1–0.3 small, 0.3–0.5 medium, ≥ 0.5 large.
- Cramér's V is the chi-square effect size on $[0, 1]$: $V = \sqrt{\chi^2 / (N(k - 1))}$. Conventional bands match Kendall's W .
- ε^2 (epsilon squared) is the Kruskal–Wallis effect size on $[0, 1]$: $\varepsilon^2 = (H - k + 1) / (N - k)$.
- Spearman's ρ is the rank correlation coefficient on $[-1, +1]$, used for the confidence–quality analyses; magnitudes follow the standard correlation conventions (small $|\rho| \approx 0.1$, medium ≈ 0.3 , large ≥ 0.5).

Bonferroni correction. A simple family-wise error-rate adjustment that divides the per-test significance threshold by the number of comparisons in the family. With m comparisons and a desired family-wise $\alpha = 0.05$, each individual test is evaluated against $\alpha' = \alpha / m$. With the three pairwise mode comparisons used throughout this work, $\alpha' = 0.0167$.

4 Evaluation and Results

This chapter reports the empirical findings of the user study and the subsequent document-quality analysis. The presentation follows the analysis plan laid out in Chapter 3: results are organised first around the per-mode self-report instruments (System Usability Scale, NASA Raw Task Load Index, Perceived AI Support), then the post-experiment mode preference ranking together with the open-ended free-text comments, and then the document-level quality measures (reference-based NLP metrics computed at the document level and the chapter level, followed by the LLM-as-a-judge evaluation). The chapter closes with a short overall summary and a descriptive Demographics section that profiles the participant sample. For every quantitative measure the per-mode results are reported in the order Form, Ask, Chat, followed by the aggregated value across all three modes; relative direction (increase, decrease, or comparable level) is stated explicitly. All inferential test statistics referenced in the running text are reproduced in full in Appendix D.

The survey analyses use the matched within-subjects sample of $n = 69$ participants. The document-quality analyses use the full set of 113 retained documents (Form $n = 37$, Ask $n = 37$, Chat $n = 39$) for descriptive statistics, and the strict matched-triple subset of $n = 26$ participants for within-subjects significance testing.

4.1 Usability — System Usability Scale

The System Usability Scale is administered after each interaction mode. It consists of ten Likert items on a five-point scale, with odd-numbered items phrased positively (higher response equals more positive evaluation) and even-numbered items phrased negatively (lower response equals more positive evaluation). The aggregated “All” column in the tables below pools the three per-mode samples into a single per-item descriptive across all 207 ratings.

4.1.1 Per-Item Results

Table 4.1 reports the mean response per item per mode and the aggregated mean across all three modes. The aggregated column is the unweighted mean of the three per-mode means.

Table 4.1: System Usability Scale, mean response per item by mode ($n = 69$ per mode). Items are reported on the original five-point Likert scale; the “All” column averages the three per-mode means. Odd items: higher is better; even items: lower is better.

#	Item	Form	Ask	Chat	All
1	I think that I would like to use this system frequently.	3.52	3.22	3.22	3.32
2	I found the system unnecessarily complex.	1.64	2.25	2.12	2.00
3	I thought the system was easy to use.	4.38	4.01	3.99	4.13
4	I think that I would need the support of a technical person to be able to use this system.	1.52	1.49	1.55	1.52
5	I found the various functions in this system were well integrated.	4.00	3.67	3.64	3.77
6	I thought there was too much inconsistency in this system.	2.16	2.04	2.01	2.07
7	I would imagine that most people would learn to use this system very quickly.	4.30	4.00	4.03	4.11
8	I found the system very cumbersome to use.	1.65	2.16	2.29	2.03
9	I felt very confident using the system.	3.83	3.84	3.90	3.86
10	I needed to learn a lot of things before I could get going with this system.	1.61	1.61	1.62	1.61

For “*I think that I would like to use this system frequently*” (item 1), Form yields the highest mean (3.52), followed by Ask and Chat at the same level (3.22 each); the aggregated mean is 3.32. Frequent-use intent therefore decreases from Form to the two LLM-driven modes by approximately 0.30 points.

For “*I found the system unnecessarily complex*” (item 2, lower is better), Form is rated lowest (1.64), Chat second (2.12), and Ask highest at 2.25; the aggregated mean is 2.00. Perceived complexity is approximately 0.5 points higher in the LLM-driven modes than in Form.

For “*I thought the system was easy to use*” (item 3), Form is rated highest (4.38), with Ask and Chat closely matched at 4.01 and 3.99; the aggregated mean is 4.13.

Ease-of-use ratings decrease by roughly 0.4 points from Form to the LLM-driven modes.

For *“I think that I would need the support of a technical person to be able to use this system”* (item 4, lower is better), the three modes are essentially indistinguishable (Form 1.52, Ask 1.49, Chat 1.55), with an aggregated mean of 1.52. None of the three modes is perceived to require technical support.

For *“I found the various functions in this system were well integrated”* (item 5), Form scores highest (4.00), Ask and Chat lower (3.67 and 3.64); aggregated mean is 3.77. The integration rating drops by approximately 0.35 points from Form to the LLM-driven modes.

For *“I thought there was too much inconsistency in this system”* (item 6, lower is better), the three modes are similar (Form 2.16, Ask 2.04, Chat 2.01), aggregated mean 2.07. Form is rated marginally higher on this negative item than the two LLM-driven modes.

For *“I would imagine that most people would learn to use this system very quickly”* (item 7), Form scores 4.30, Ask 4.00, Chat 4.03; aggregated mean is 4.11. Perceived learnability decreases by approximately 0.30 points from Form to the LLM-driven modes.

For *“I found the system very cumbersome to use”* (item 8, lower is better), the largest spread is observed: Form 1.65, Ask 2.16, Chat 2.29; aggregated mean 2.03. Cumbersomeness is rated approximately 0.5–0.65 points higher in the LLM-driven modes than in Form.

For *“I felt very confident using the system”* (item 9), all three modes are rated similarly and slightly higher in the LLM-driven modes (Form 3.83, Ask 3.84, Chat 3.90); aggregated mean 3.86. Within-system confidence is broadly comparable across modes, with a small upward trend from Form to Chat.

For *“I needed to learn a lot of things before I could get going with this system”* (item 10, lower is better), the three modes are virtually identical (Form 1.61, Ask 1.61, Chat 1.62), aggregated mean 1.61. The amount of prior learning required is perceived as low and equal across all modes.

Table 4.2 reports the full descriptive distribution per item per mode (mean, standard deviation, median, Q1, Q3). The minimum and maximum are 1 and 5 for every

4 Evaluation and Results

item $\times$ mode combination and are omitted for compactness.

The per-item distributions are visualised in Figure 4.1.

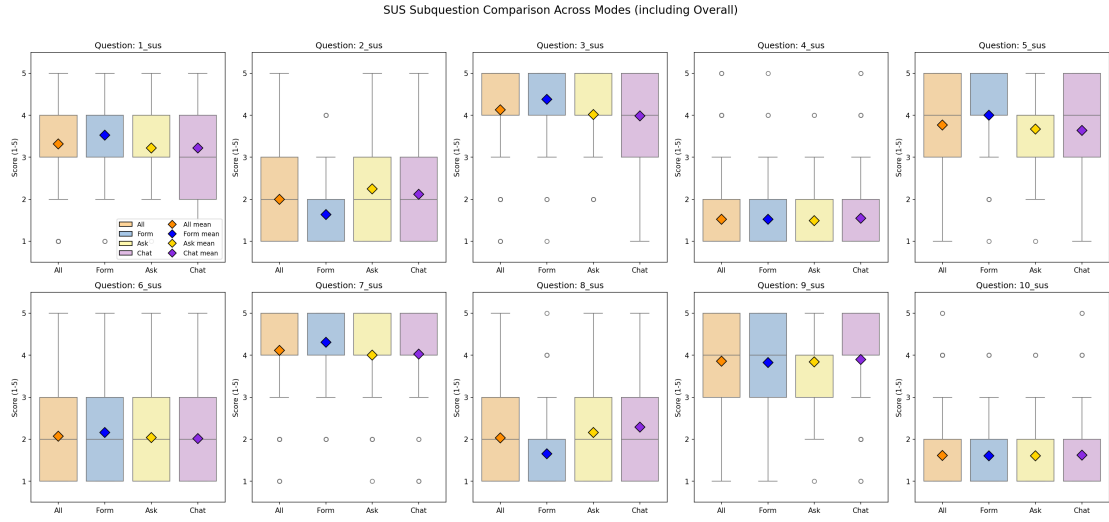


Figure 4.1: Box plots of the ten SUS items by mode ($n = 69$ per mode). Boxes show the interquartile range; the central line marks the median, and the diamond marks the mean.

A Friedman omnibus test on the per-mode aggregate yields $\chi^2(2) = 5.51$, $p = 0.064$, $W = 0.040$, which does not reach the $\alpha = 0.05$ threshold. Pairwise Wilcoxon signed-rank post-hoc tests with Bonferroni correction ($\alpha = 0.0167$) indicate a significant Form-vs-Ask difference ($W = 578.5$, $p = 0.013$, rank-biserial $r = +0.368$, medium effect), a non-significant Form-vs-Chat difference ($p = 0.018$, just above the Bonferroni threshold), and no Ask-vs-Chat difference ($p = 0.666$). Full test statistics are reproduced in Appendix D.

4.2 Cognitive Load — NASA Raw Task Load Index

The Raw Task Load Index captures six dimensions of perceived workload — Mental Demand, Physical Demand, Temporal Demand, Performance, Effort, and Frustration — each on a five-point Likert scale. Performance is reported in the positive-coded direction (higher equals worse perceived performance) for internal consistency with the other subscales. The overall R-TLX in this study is the unweighted mean of the six subscales, rescaled to a 1–21 range.

Table 4.2: System Usability Scale, full per-item descriptive distribution by mode
($n = 69$ per mode). Source: sus_per_item_summary.csv.

#	Mode	Mean	SD	Median	Q1	Q3
1	All	3.32	1.14	3	3	4
	Form	3.52	1.09	4	3	4
	Ask	3.22	1.08	3	3	4
	Chat	3.22	1.24	3	2	4
2	All	2.00	1.12	2	1	3
	Form	1.64	0.92	1	1	2
	Ask	2.25	1.22	2	1	3
	Chat	2.12	1.12	2	1	3
3	All	4.13	0.97	4	4	5
	Form	4.38	0.91	5	4	5
	Ask	4.01	0.81	4	4	5
	Chat	3.99	1.13	4	3	5
4	All	1.52	0.89	1	1	2
	Form	1.52	0.85	1	1	2
	Ask	1.49	0.80	1	1	2
	Chat	1.55	1.01	1	1	2
5	All	3.77	1.05	4	3	5
	Form	4.00	0.92	4	4	5
	Ask	3.67	1.02	4	3	4
	Chat	3.64	1.16	4	3	5
6	All	2.07	1.00	2	1	3
	Form	2.16	0.96	2	1	3
	Ask	2.04	1.06	2	1	3
	Chat	2.01	0.98	2	1	3
7	All	4.11	0.98	4	4	5
	Form	4.30	0.96	5	4	5
	Ask	4.00	0.95	4	4	5
	Chat	4.03	1.01	4	4	5
8	All	2.03	1.13	2	1	3
	Form	1.65	0.94	1	1	2
	Ask	2.16	1.12	2	1	3
	Chat	2.29	1.23	2	1	3
9	All	3.86	1.04	4	3	5
	Form	3.83	1.04	4	3	5
	Ask	3.84	0.93	4	3	4
	Chat	3.90	1.15	4	4	5
10	All	1.61	0.90	1	1	2
	Form	1.61	0.86	1	1	2
	Ask	1.61	0.81	1	1	2
	Chat	1.62	0.83	1	1	2

4.2.1 Per-Subscale Results

Table 4.3 reports the per-subscale mean and standard deviation per mode and the aggregated mean across modes. Figure 4.2 shows the corresponding distributions.

Table 4.3: NASA R-TLX subscale means $\pm$ standard deviation by mode ($n = 69$ per mode), 1–21 scale. The aggregated “All” column is the unweighted mean of the three per-mode means. All subscales are coded so that lower equals less workload.

Subscale	Form	Ask	Chat	All
Mental Demand	5.77 ± 4.23	6.80 ± 4.60	6.36 ± 5.27	6.31
Physical Demand	2.77 ± 2.69	3.54 ± 3.89	3.26 ± 3.43	3.19
Temporal Demand	4.16 ± 3.79	4.97 ± 4.12	5.54 ± 4.74	4.89
Effort	5.68 ± 4.53	6.39 ± 4.37	6.32 ± 4.74	6.13
Frustration	5.88 ± 4.80	7.58 ± 5.30	7.13 ± 5.41	6.86
Performance (pos.)	8.71 ± 5.44	8.68 ± 5.60	9.39 ± 6.32	8.93

Table 4.4 reports the full descriptive distribution per subscale per mode (mean, standard deviation, median, Q1, Q3, min, max). All subscales use the 1–21 scale where lower equals less workload (Performance is positive-coded for internal consistency).

For *Mental Demand*, Form scores lowest (mean 5.77), Chat next (6.36), and Ask highest (6.80); the aggregated mean is 6.31. Mental Demand increases by approximately 0.6–1.0 points when moving from Form to the LLM-driven modes.

For *Physical Demand*, Form scores lowest (2.77), Chat second (3.26), Ask highest (3.54); aggregated 3.19. The absolute spread is small (≈ 0.8 points). Physical Demand is the lowest-rated subscale in all three modes.

For *Temporal Demand*, the values increase monotonically from Form (4.16) to Ask (4.97) to Chat (5.54); aggregated 4.89. Temporal Demand is approximately 1.4 points higher in Chat than in Form, the largest monotonic increase observed across the six subscales.

For *Effort*, Form scores lowest (5.68), with Ask (6.39) and Chat (6.32) at a closely comparable level; aggregated 6.13. Effort increases by approximately 0.65 points from Form to the LLM-driven modes.

For *Frustration*, Form scores lowest (5.88), Chat second (7.13), and Ask highest

Table 4.4: NASA R-TLX, full per-subscale descriptive distribution by mode ($n = 69$ per mode), 1–21 scale. Source: `tlx_per_item_summary.csv`.

Subscale	Mode	Mean	SD	Median	Q1	Q3	Min	Max
Mental Demand	All	6.31	4.75	5.0	2.0	9.5	1	21
	Form	5.77	4.26	5.0	2.0	9.0	1	17
	Ask	6.80	4.63	6.0	3.0	10.0	1	21
	Chat	6.36	5.31	5.0	2.0	10.0	1	21
Physical Demand	All	3.19	3.40	2.0	1.0	4.0	1	21
	Form	2.77	2.71	1.0	1.0	4.0	1	14
	Ask	3.54	3.92	2.0	1.0	5.0	1	21
	Chat	3.26	3.46	2.0	1.0	4.0	1	20
Temporal Demand	All	4.89	4.28	4.0	1.0	7.0	1	19
	Form	4.16	3.82	3.0	1.0	6.0	1	15
	Ask	4.97	4.15	4.0	1.0	7.0	1	18
	Chat	5.54	4.78	4.0	2.0	8.0	1	19
Effort	All	6.13	4.57	5.0	3.0	10.0	1	20
	Form	5.68	4.56	4.0	2.0	9.0	1	17
	Ask	6.39	4.41	5.0	3.0	10.0	1	19
	Chat	6.32	4.77	5.0	3.0	10.0	1	20
Frustration	All	6.86	5.24	5.0	2.5	11.0	1	21
	Form	5.88	4.84	5.0	2.0	8.0	1	18
	Ask	7.58	5.34	6.0	3.0	12.0	1	21
	Chat	7.13	5.44	5.0	3.0	11.0	1	19
Performance (pos.)	All	8.93	5.82	9.0	4.0	13.0	1	21
	Form	8.71	5.48	8.0	4.0	12.0	1	21
	Ask	8.68	5.64	8.0	4.0	12.0	1	19
	Chat	9.39	6.36	10.0	3.0	14.0	1	21

4 Evaluation and Results

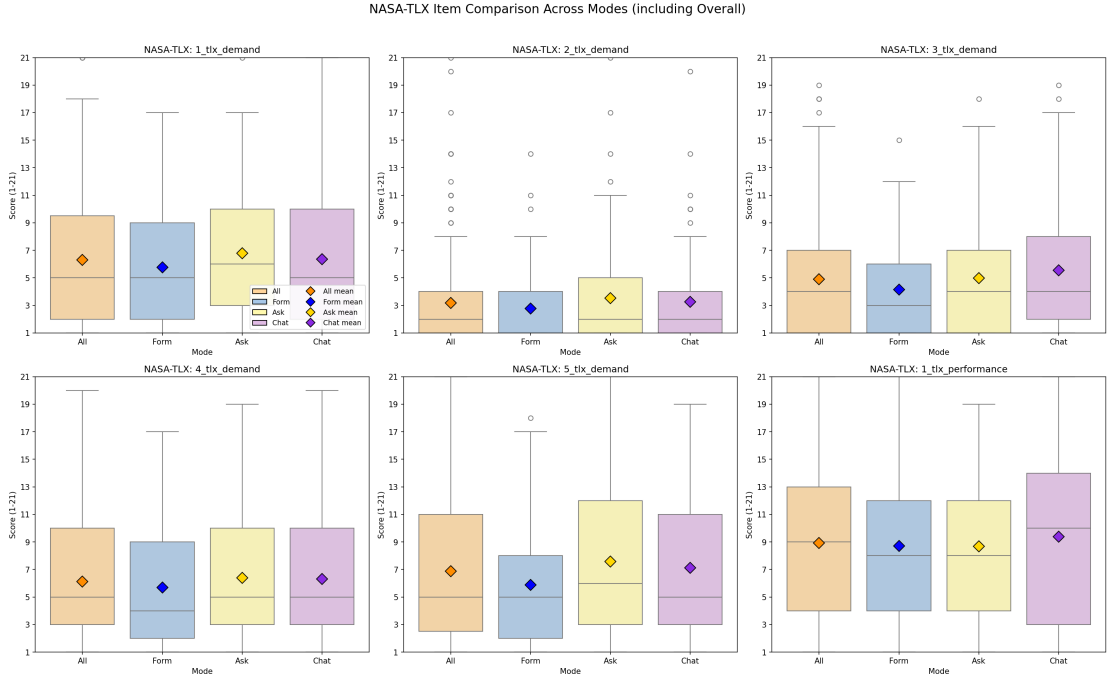


Figure 4.2: Box plots of the six R-TLX subscales by mode ($n = 69$ per mode).

(7.58); aggregated 6.86. Frustration shows the largest absolute gap between Form and the LLM-driven modes (+1.25 to +1.70 points).

For *Performance* (positive-coded, higher equals worse perceived performance), Form and Ask are essentially identical (8.71 and 8.68); Chat is slightly higher at 9.39; aggregated 8.93. Perceived self-performance is comparable across modes, with a marginal degradation in Chat.

A Friedman omnibus across the six subscales is non-significant ($\chi^2(2) = 5.24$, $p = 0.073$, $W = 0.038$). Among the six subscales, two reach significance: Temporal Demand ($\chi^2(2) = 12.77$, $p = 0.002$, $W = 0.093$) and Frustration ($\chi^2(2) = 8.93$, $p = 0.012$, $W = 0.065$). Pairwise post-hoc Wilcoxon signed-rank tests with Bonferroni correction identify Form-vs-Ask and Form-vs-Chat as significant on these two subscales, with medium-to-large rank-biserial effect sizes ($|r| = 0.36$ to 0.57); Ask versus Chat is non-significant on every measure. Full test statistics are reproduced in Appendix D.

4.3 Perceived AI Support — AI Questionnaire

After each interaction mode, in addition to the SUS and R-TLX, participants completed a six-item Perceived AI Support questionnaire on a five-point Likert scale (1 = strong disagreement, 5 = strong agreement). The six items capture distinct facets of how participants perceived the AI's contribution to the document they had just created: helpfulness (item 1), efficiency (item 2), transparency (item 3), trust (item 4), legal-basis confidence (item 5), and re-use intent (item 6). Item 5 is the primary self-report confidence measure referenced in Chapter 3. The aggregated “All” column in the tables below pools the three per-mode samples into a single per-item descriptive across all 207 ratings.

4.3.1 Per-Item Means

Table 4.5 reports the mean response per item per mode and the aggregated mean across the three modes; the full descriptive distribution per item is reported in Table 4.6. Figure 4.3 shows the corresponding box-plot distributions.

Table 4.5: Perceived AI Support, mean response per item by mode ($n = 69$ per mode), five-point Likert scale (higher is better). The aggregated “All” column is the unweighted mean of the three per-mode means.

#	Item	Form	Ask	Chat	All
1	The AI was helpful in creating this document.	4.23	4.32	4.13	4.23
2	The AI helped me to complete the document faster.	4.43	4.25	3.99	4.22
3	I understood why the AI suggested specific GDPR fields.	3.51	4.16	3.75	3.81
4	I trust that the AI made correct suggestions.	3.41	3.80	3.68	3.63
5	I am confident that the AI identified the correct legal basis.	3.58	3.74	3.81	3.71
6	I would use this mode again.	3.99	3.55	3.42	3.65

For “*The AI was helpful in creating this document*” (item 1), Form scores 4.23, Ask 4.32, Chat 4.13; aggregated 4.23. Helpfulness is rated highest in Ask and lowest in Chat, with a small absolute spread of roughly 0.20 points.

For “*The AI helped me to complete the document faster*” (item 2), Form is rated highest (4.43), Ask second (4.25), Chat lowest (3.99); aggregated 4.22. Perceived

4 Evaluation and Results

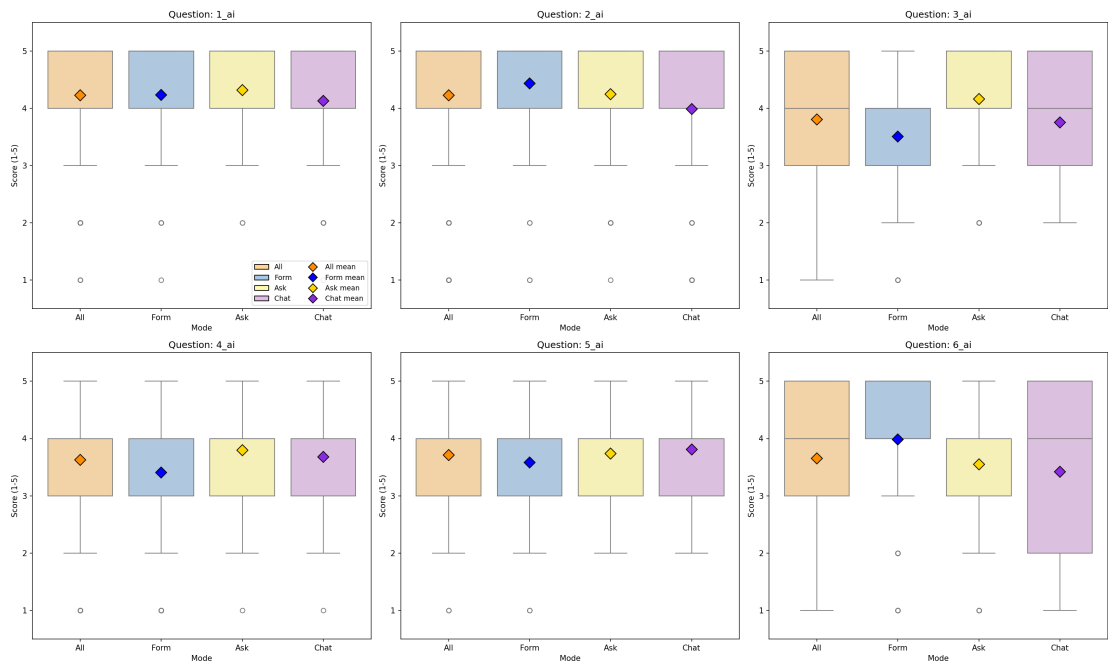


Figure 4.3: Box plots of the six Perceived AI Support items by mode ($n = 69$ per mode). Boxes show the interquartile range; the central line marks the median, and the diamond marks the mean.

speed-up decreases monotonically with increasing LLM involvement, dropping by approximately 0.45 points from Form to Chat.

For *“I understood why the AI suggested specific GDPR fields”* (item 3), Form scores lowest (3.51), Chat second (3.75), Ask highest (4.16); aggregated 3.81. Transparency is rated approximately 0.65 points higher in Ask than in Form. Ask is the only mode that explicitly explains each suggestion.

For *“I trust that the AI made correct suggestions”* (item 4), Form scores lowest (3.41), Chat second (3.68), Ask highest (3.80); aggregated 3.63. Trust increases by approximately 0.27–0.39 points from Form to the LLM-driven modes.

For *“I am confident that the AI identified the correct legal basis”* (item 5, primary confidence measure), the three modes show a monotonic increase: Form 3.58, Ask 3.74, Chat 3.81; aggregated 3.71. Confidence in legal correctness rises by approximately 0.23 points from Form to Chat.

For *“I would use this mode again”* (item 6), Form scores highest (3.99), Ask second (3.55), Chat lowest (3.42); aggregated 3.65. Re-use intent decreases by approximately 0.57 points from Form to Chat, the largest negative gap of the six items.

4.3.2 Per-Item Descriptive Distributions

Table 4.6 reports the full descriptive distribution per item per mode: mean, standard deviation, median, Q1, and Q3. The minimum is 1 and the maximum is 5 for every cell (both endpoints of the Likert scale are realised in every item $\times$ mode combination), so they are omitted from the table for compactness.

An omnibus Friedman test on the per-mode aggregate is non-significant ($\chi^2(2) = 1.05$, $p = 0.592$, $W = 0.008$), indicating that perceived AI support does not differ meaningfully across modes when the six items are considered together. The single legal-basis confidence item 5_{ai} is also non-significant under Friedman ($\chi^2(2) = 3.76$, $p = 0.152$). The re-use item 6_{ai} does reach significance ($\chi^2(2) = 11.12$, $p = 0.004$, $W = 0.081$), with a large rank-biserial effect for Form versus Chat ($r = +0.543$). Full statistics are reproduced in Appendix D.

Table 4.6: Perceived AI Support, full per-item descriptive distribution by mode ($n = 69$ per mode). Source: ai_support_per_item_summary.csv.

#	Mode	Mean	SD	Median	Q1	Q3
1	All	4.23	0.91	4	4	5
	Form	4.23	0.97	5	4	5
	Ask	4.32	0.76	4	4	5
	Chat	4.13	1.00	4	4	5
2	All	4.22	1.13	5	4	5
	Form	4.43	0.96	5	4	5
	Ask	4.25	1.03	5	4	5
	Chat	3.99	1.32	4	4	5
3	All	3.81	1.06	4	3	5
	Form	3.51	1.17	4	3	4
	Ask	4.16	0.92	4	4	5
	Chat	3.75	0.98	4	3	5
4	All	3.63	0.97	4	3	4
	Form	3.41	1.09	4	3	4
	Ask	3.80	0.87	4	3	4
	Chat	3.68	0.92	4	3	4
5	All	3.71	0.88	4	3	4
	Form	3.58	0.90	4	3	4
	Ask	3.74	0.87	4	3	4
	Chat	3.81	0.88	4	3	4
6	All	3.65	1.25	4	3	5
	Form	3.99	1.14	4	4	5
	Ask	3.55	1.16	4	3	4
	Chat	3.42	1.39	4	2	5

4.4 Mode Preference and Free-Text Feedback

After all three interaction modes had been completed, participants provided two final inputs: an overall ranking of the three modes from most to least preferred, and an open-ended summary comment on the study experience. In addition, after each individual mode, an optional per-mode comment was collected. This section reports the ranking distribution, the inferential test on the rank-1 votes, and a thematic summary of all 88 free-text comments contributed by 46 unique participants.

4.4.1 Mode Preference Ranking

Table 4.7 summarises the post-experiment ranking. Participants assigned each of the three modes a rank from 1 (most preferred) to 3 (least preferred); “Rank-1 votes” counts the participants who placed the mode first.

Table 4.7: Post-experiment mode preference ranking ($n = 69$). “Rank-1 votes” counts the number of participants who placed the mode first; “Share” is the corresponding percentage; “Mean rank” averages the assigned 1–3 rank across all 69 participants.

Mode	Rank-1 votes	Share	Mean rank
Form	37	53.6 %	1.78
Chat	20	29.0 %	2.07
Ask	12	17.4 %	2.14

Form receives the plurality of first-place votes (53.6 %), followed by Chat (29.0 %) and Ask (17.4 %). The mean ranks follow the same ordering. A chi-square goodness-of-fit test against a uniform null rejects the null hypothesis ($\chi^2(2, N = 69) = 14.17$, $p < 0.001$, Cramér’s $V = 0.320$, medium effect). The complementary Friedman test on the full rank distribution is non-significant ($\chi^2(2) = 5.07$, $p = 0.079$), reflecting that the strong Form plurality at rank 1 is partially balanced by variance at ranks 2 and 3. Full statistics are reproduced in Appendix D.

The complete distribution of all three rank positions across the three modes is visualised in Figure 4.4.

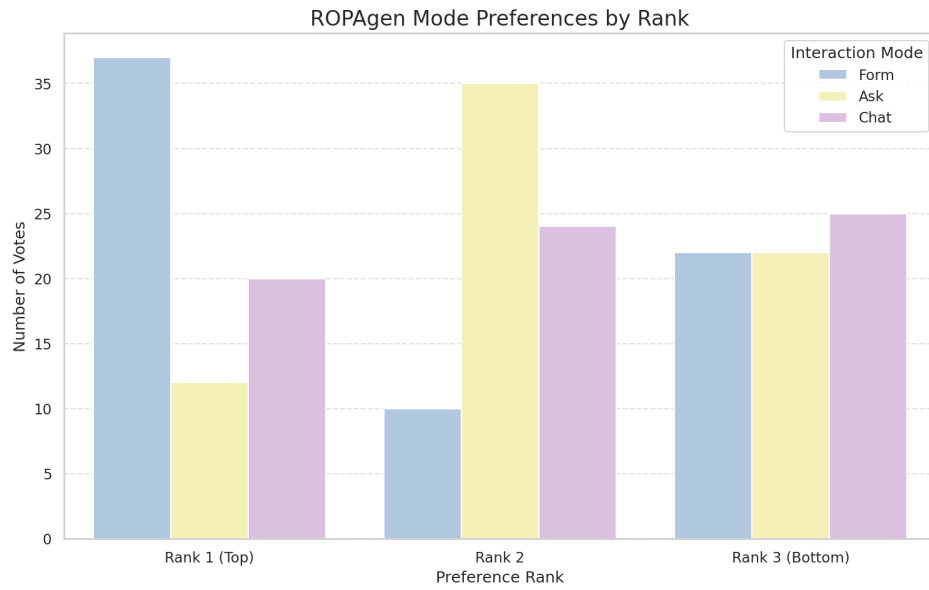


Figure 4.4: Distribution of all three rank positions per mode ($n = 69$).

4.4.2 Free-Text Comments

Each mode collected an optional per-mode open-ended comment, and after all three modes participants wrote an overall summary comment. A total of 88 comments were recorded across 46 unique participants. Comments are written in German with a small minority in English; they are summarised thematically below. The presentation preserves the order Form, Ask, Chat, and concludes with the across-modes overall comments.

Form. Comments on Form cluster around speed, clarity, and control. Recurring observations describe the mode as the fastest of the three, the easiest to obtain an overview of, and the most “integrated” into the document creation workflow because the checkbox interface makes the selection space visible at a glance. Several participants emphasise that Form “takes the most work off you” (“*nimmt einem die meiste Arbeit ab*”) by pre-populating the option set, while a smaller subset notes the opposite concern: that the absence of an interactive AI commentary makes it unclear on what basis the model produces its suggestions, and that participants would not feel safe relying on Form output without an additional verification step. Several comments describe specific UX issues: long text fields rendered as a sin-

gle narrow line on mobile, the inability to provide the AI with additional context for its suggestions, and the impression that the AI “guesses” without scenario-specific input.

Ask. Comments on Ask are the most polarised of the three modes. A consistent theme is that Ask is the most cumbersome mode in terms of workflow: the requirement to manually copy the AI’s suggested text into the form field is repeatedly described as redundant work, particularly because participants reported that the model often produced suggestions in a free-text style that did not match the structured form fields. Several participants observed that the lack of a persistent context across fields forced them to re-enter the same information repeatedly. Conversely, explanatory text generated by Ask is consistently described as the most informative of the three modes, supporting the high transparency rating visible in item 3 of the AI Support questionnaire. A frequent suggestion is to combine Ask’s explanations with Form’s structured field layout.

Chat. Comments on Chat are dominated by two themes: comfort and opacity. On the comfort side, several participants describe Chat as the most intuitive of the three modes, especially because the AI takes over the work of mapping the conversation onto the structured ROPA fields, leaving the participant only with the task of describing the processing scenario. On the opacity side, an equally large group of participants reports uncertainty about what is actually being written into the document — whether the entire chat history is used for generation, whether each individual message must be confirmed, and what the document looks like during the conversation. Specific defects mentioned include the AI occasionally switching from German to English mid-response, occasional emission of raw `DATA_UPDATE` tokens, and a slow “streaming” rendering style that participants experienced as artificially throttled.

Across modes (overall comments). The overall comments recapitulate the per-mode pattern. A clear majority describes Form as the fastest and most ergonomic mode, particularly for participants who feel they “do not know enough” to interrogate the model meaningfully. Chat is typically placed second and motivated by the

impression that “the AI takes care of it,” which several participants explicitly note increases their *subjective* sense of safety even though they cannot verify the output. Ask is most often placed third, with the manual copy-paste step cited as the principal reason. A recurring constructive suggestion across participants is a hybrid mode that combines the visible option set of Form with the conversational, context-aware behaviour of Chat or Ask. Several participants explicitly add a caveat that they are novices in the legal domain and that their preference ordering might shift if they had prior GDPR knowledge.

4.5 Document Quality — Reference-Based NLP Metrics

Document-quality results are reported on the full retained set of 113 documents (Form $n = 37$, Ask $n = 37$, Chat $n = 39$) for descriptive statistics, and on the strict matched-triple subset of $n = 26$ for within-subjects significance tests. All metrics are computed against the single expert-authored Form-mode reference document described in Chapter 3. For each metric, the per-mode results are reported in the order Form, Ask, Chat, followed by the aggregated value across all three modes. The descriptive tables in this section report — per metric and per mode — the full empirical distribution (n , min, Q1, median, Q3, max, mean, SD) computed from the raw per-document scores in the pipeline outputs `document_results.csv` and `chapter_results.csv`.

4.5.1 Document-Level Results

Lexical Metrics

Table 4.8 reports the full distribution of the five lexical metrics (BLEU, ROUGE-1/2/L, METEOR) per mode, plus the aggregated row across all 113 documents. Figure 4.5 shows the corresponding box-plot distributions.

Table 4.8: Document-level lexical metric distributions by mode (full sample $n = 113$). Source: doc_nlp_summary.csv.

Metric	Mode	n	Min	Q1	Median	Q3	Max	Mean	SD
BLEU	Form	37	0.064	0.088	0.124	0.158	0.230	0.129	0.048
	Ask	37	0.047	0.083	0.125	0.148	0.246	0.127	0.056
	Chat	39	0.067	0.122	0.134	0.163	0.235	0.143	0.039
	All	113	0.047	0.094	0.128	0.159	0.246	0.133	0.048
ROUGE-1	Form	37	0.313	0.421	0.493	0.569	0.644	0.488	0.098
	Ask	37	0.322	0.412	0.449	0.522	0.618	0.465	0.088
	Chat	39	0.330	0.468	0.517	0.561	0.645	0.511	0.071
	All	113	0.313	0.425	0.491	0.560	0.645	0.488	0.087
ROUGE-2	Form	37	0.178	0.241	0.301	0.361	0.455	0.306	0.076
	Ask	37	0.121	0.200	0.238	0.275	0.429	0.257	0.080
	Chat	39	0.131	0.253	0.283	0.344	0.447	0.291	0.069
	All	113	0.121	0.234	0.268	0.344	0.455	0.285	0.077
ROUGE-L	Form	37	0.238	0.320	0.391	0.468	0.566	0.388	0.094
	Ask	37	0.194	0.296	0.326	0.376	0.545	0.347	0.091
	Chat	39	0.230	0.348	0.387	0.444	0.544	0.389	0.072
	All	113	0.194	0.311	0.359	0.441	0.566	0.375	0.087
METEOR	Form	37	0.335	0.367	0.419	0.442	0.514	0.412	0.051
	Ask	37	0.234	0.327	0.366	0.409	0.488	0.365	0.068
	Chat	39	0.235	0.374	0.389	0.421	0.501	0.390	0.057
	All	113	0.234	0.359	0.388	0.431	0.514	0.389	0.062

4 Evaluation and Results

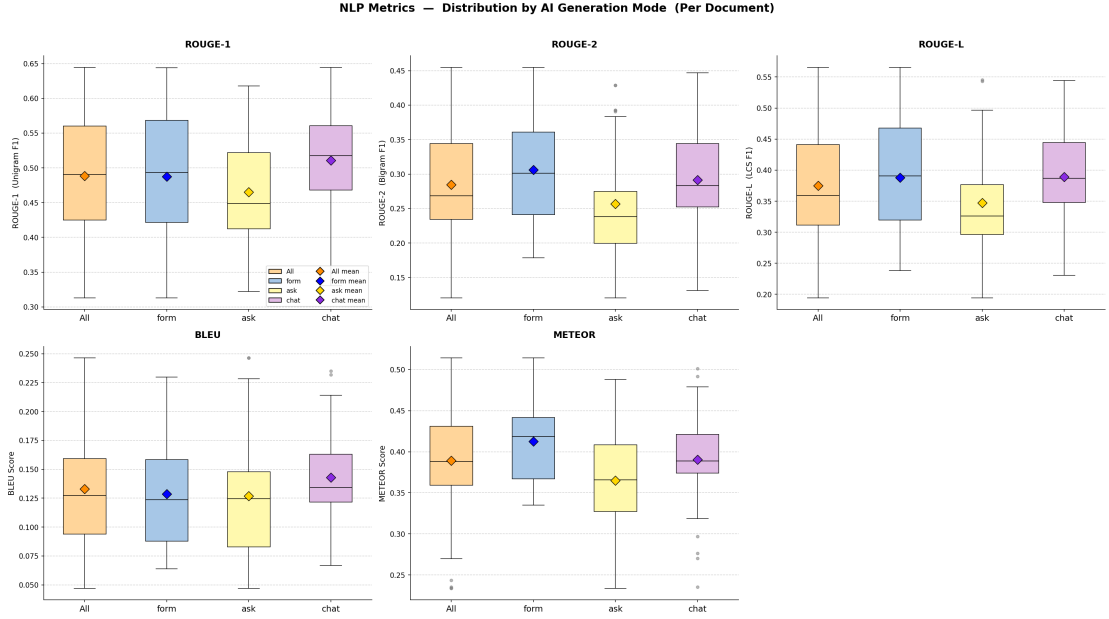


Figure 4.5: Document-level distributions of BLEU, ROUGE-1/2/L, and METEOR by mode (full sample $n = 113$). Boxes show the interquartile range; the diamond marks the mean.

BLEU. BLEU (*Bilingual Evaluation Understudy*) [10] measures the geometric mean of n -gram precisions between a candidate and one or more reference texts, multiplied by a brevity penalty that discourages outputs much shorter than the reference. With p_n the modified n -gram precision, $w_n = 1/N$ uniform weights, c the candidate length, and r the closest reference length,

$$\text{BLEU} = \text{BP} \cdot \exp\left(\sum_{n=1}^N w_n \log p_n\right), \quad \text{BP} = \begin{cases} 1 & \text{if } c > r \\ e^{1-r/c} & \text{if } c \leq r. \end{cases} \quad (4.1)$$

This work uses $N = 4$ (matching unigrams through 4-grams). BLEU is a precision-oriented surface-form metric: it rewards exact n -gram overlap and is therefore harsh on paraphrase.

For *BLEU*, Form yields a mean of 0.129, Ask 0.127, Chat 0.143; the aggregated mean is 0.133. Chat scores marginally higher than Form and Ask, which are essentially indistinguishable. The Friedman omnibus on the matched subset is significant ($\chi^2(2) = 6.23$, $p = 0.044$); the significant pairwise difference is Chat $>$ Ask ($r = -0.578$, large effect).

ROUGE-1, ROUGE-2, ROUGE-L. ROUGE (*Recall-Oriented Understudy for Gisting Evaluation*) [6] reports the F1 score of n -gram overlap between candidate and reference. ROUGE-1 counts unigrams, ROUGE-2 counts bigrams, and ROUGE-L counts the length of the longest common subsequence (LCS). Let X be the candidate, Y the reference, and $\text{LCS}(X, Y)$ their longest common subsequence; then for ROUGE-N

$$P_N = \frac{|\text{ngrams}(X) \cap \text{ngrams}(Y)|}{|\text{ngrams}(X)|}, \quad R_N = \frac{|\text{ngrams}(X) \cap \text{ngrams}(Y)|}{|\text{ngrams}(Y)|}, \quad (4.2)$$

and for ROUGE-L

$$P_L = \frac{|\text{LCS}(X, Y)|}{|X|}, \quad R_L = \frac{|\text{LCS}(X, Y)|}{|Y|}. \quad (4.3)$$

The reported value in both cases is the harmonic mean (F1) of precision and recall. ROUGE rewards lexical overlap with the reference and is more permissive than BLEU in word order.

For *ROUGE-1*, Form yields 0.488, Ask 0.465, Chat 0.511; aggregated 0.488. ROUGE-1 increases from Ask to Form to Chat. The Friedman omnibus is significant ($\chi^2(2) = 10.69$, $p = 0.005$); Form > Ask and Chat > Ask are significant pairwise differences with large effects, while Form versus Chat is not significant.

For *ROUGE-2*, Form yields the highest mean (0.306), Chat second (0.291), Ask lowest (0.257); aggregated 0.285. ROUGE-2 follows the Form > Chat > Ask ordering. The Friedman omnibus is significant ($\chi^2(2) = 13.46$, $p = 0.001$); Form > Ask is significant ($r = +0.829$, large effect), as is Chat > Ask.

For *ROUGE-L*, Form (0.388) and Chat (0.389) are essentially tied, with Ask trailing (0.347); aggregated 0.375. The Friedman omnibus is significant ($\chi^2(2) = 11.31$, $p = 0.004$); Form > Ask and Chat > Ask are significant pairwise differences with large effects.

METEOR. METEOR (*Metric for Evaluation of Translation with Explicit ORdering*) [2] is the weighted harmonic mean of unigram precision P and recall R , with recall up-weighted by a factor α , multiplied by a fragmentation penalty that grows with the number of disjoint matching chunks ch relative to the number of matched unigrams

m :

$$\text{METEOR} = \underbrace{\frac{P \cdot R}{\alpha P + (1 - \alpha) R}}_{F_{\text{mean}}} \cdot \underbrace{\left(1 - \gamma \left(\frac{\text{ch}}{m}\right)^\beta\right)}_{\text{penalty}}. \quad (4.4)$$

Standard values $\alpha = 0.9$, $\beta = 3$, $\gamma = 0.5$ are used; matching is performed against exact, stemmed, and synonym/paraphrase matches via WordNet. METEOR therefore rewards both lexical recall and contiguous matching.

For *METEOR*, Form yields the highest mean (0.412), Chat second (0.390), Ask lowest (0.365); aggregated 0.389. The Friedman omnibus is significant ($\chi^2(2) = 9.92$, $p = 0.007$); Form > Ask is significant ($r = +0.812$, large effect), and the remaining pairwise differences do not reach the Bonferroni threshold.

BERT-Based and Sentence-Embedding Metrics

Table 4.9 reports the same full distribution for the BERT-based metrics (BERTScore Precision / Recall / F1) and the document-level SBERT cosine similarity computed with ModernBERT embeddings. Figure 4.6 shows the corresponding box-plot distributions.

BERTScore Precision, Recall, F1. BERTScore [17] replaces surface-form n -gram matching with contextual token embeddings. Each token in the candidate $\hat{x}$ and the reference x is encoded with a pretrained transformer (here, ModernBERT [16] via the `bert_score` library). For each candidate token, the maximum cosine similarity with any reference token is taken, and vice versa for recall. With L2-normalised embeddings $\mathbf{x}_i$ and $\hat{\mathbf{x}}_j$,

$$P_{\text{BERT}} = \frac{1}{|\hat{x}|} \sum_{\hat{x}_j \in \hat{x}} \max_{x_i \in x} \mathbf{x}_i^\top \hat{\mathbf{x}}_j, \quad R_{\text{BERT}} = \frac{1}{|x|} \sum_{x_i \in x} \max_{\hat{x}_j \in \hat{x}} \mathbf{x}_i^\top \hat{\mathbf{x}}_j, \quad (4.5)$$

$$F1_{\text{BERT}} = \frac{2 \cdot P_{\text{BERT}} \cdot R_{\text{BERT}}}{P_{\text{BERT}} + R_{\text{BERT}}}. \quad (4.6)$$

Precision measures how many candidate tokens have a close match in the reference (penalising spurious content), *recall* measures how many reference tokens have a close match in the candidate (penalising omission), and *F1* is their harmonic

Table 4.9: Document-level BERT-based and SBERT distributions by mode (full sample $n = 113$). Source: doc_bert_summary.csv.

Metric	Mode	n	Min	Q1	Median	Q3	Max	Mean	SD
BERTScore Precision	Form	37	0.862	0.880	0.889	0.899	0.912	0.889	0.014
	Ask	37	0.861	0.872	0.889	0.899	0.918	0.886	0.016
	Chat	39	0.867	0.890	0.896	0.902	0.922	0.895	0.012
	All	113	0.861	0.880	0.891	0.900	0.922	0.890	0.014
BERTScore Recall	Form	37	0.914	0.919	0.923	0.926	0.936	0.923	0.005
	Ask	37	0.861	0.908	0.915	0.918	0.925	0.908	0.017
	Chat	39	0.874	0.908	0.917	0.922	0.933	0.913	0.013
	All	113	0.861	0.914	0.918	0.922	0.936	0.915	0.014
BERTScore F1	Form	37	0.887	0.900	0.906	0.912	0.920	0.905	0.009
	Ask	37	0.863	0.892	0.896	0.906	0.918	0.897	0.013
	Chat	39	0.872	0.899	0.906	0.911	0.922	0.904	0.010
	All	113	0.863	0.895	0.904	0.910	0.922	0.902	0.011
SBERT (ModernBERT)	Form	37	0.974	0.984	0.988	0.989	0.992	0.986	0.004
	Ask	37	0.971	0.983	0.987	0.990	0.993	0.986	0.005
	Chat	39	0.985	0.988	0.990	0.991	0.994	0.990	0.002
	All	113	0.971	0.985	0.988	0.990	0.994	0.987	0.004

4 Evaluation and Results

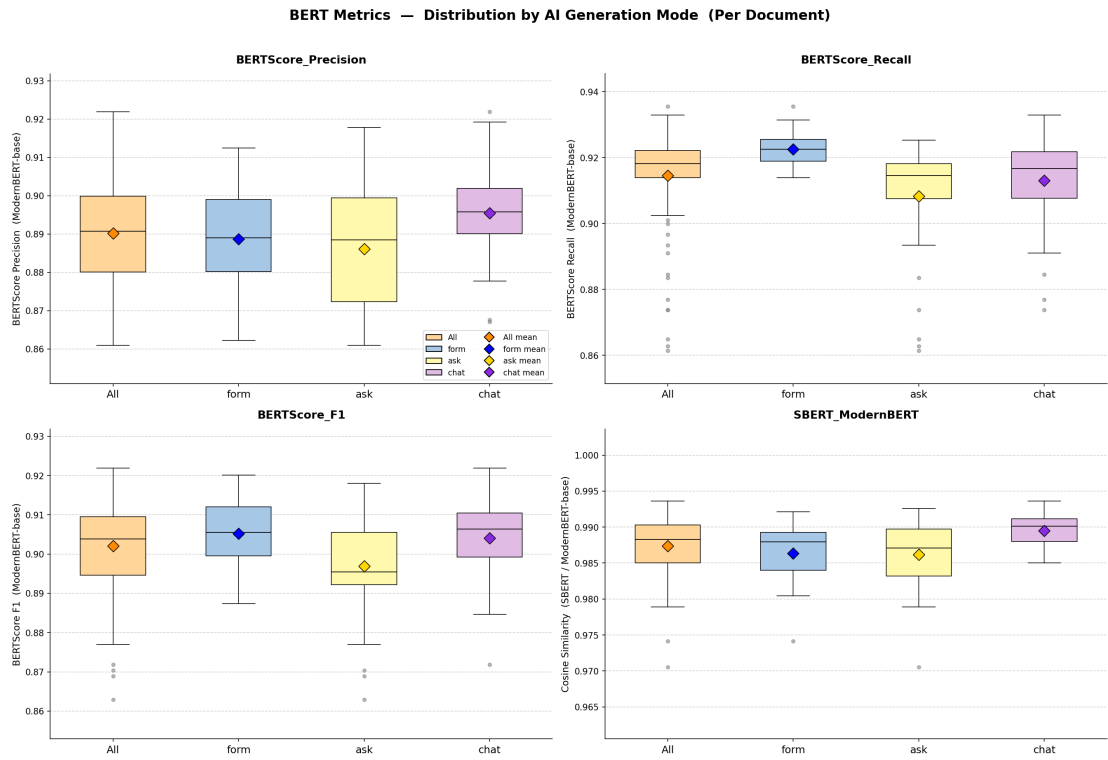


Figure 4.6: Document-level distributions of BERTScore Precision / Recall / F1 and SBERT cosine similarity (ModernBERT) by mode (full sample $n = 113$).

mean. Because the comparison is at the embedding level, BERTScore credits paraphrase that shares meaning even without surface-form overlap.

For *BERTScore Precision*, Chat yields the highest mean (0.895), Form second (0.889), Ask lowest (0.886); aggregated 0.890. The Friedman omnibus is significant ($\chi^2(2) = 18.54, p < 0.001$); Form > Ask and Chat > Ask are significant.

For *BERTScore Recall*, Form yields the highest mean (0.923), with Chat (0.913) and Ask (0.908) trailing; aggregated 0.915. The Friedman omnibus is significant ($\chi^2(2) = 30.77, p < 0.001, W = 0.410$, the largest effect observed in the metric pipeline); Form > Ask is significant with the maximum possible matched-pairs effect ($r = +1.000$), and Form > Chat is also significant ($r = +0.687$, large effect). Ask versus Chat is non-significant.

For *BERTScore F1*, Form yields the highest mean (0.905), Chat second (0.904), Ask lowest (0.897); aggregated 0.902. The Friedman omnibus is significant ($\chi^2(2) = 13.00, p = 0.002$); Form > Ask and Chat > Ask are significant with large effects. Form versus Chat is non-significant on F1, reflecting the trade-off between Form’s higher Recall and Chat’s higher Precision.

SBERT (ModernBERT) cosine similarity. Sentence-BERT [11] produces a single fixed-length embedding per document by mean-pooling token embeddings from a transformer (here, ModernBERT). Document similarity is the cosine of the angle between the two document embeddings:

$$\text{SBERT}(d_1, d_2) = \frac{\mathbf{e}(d_1)^\top \mathbf{e}(d_2)}{\|\mathbf{e}(d_1)\| \|\mathbf{e}(d_2)\|}. \quad (4.7)$$

Because the comparison is at the document-embedding level rather than the token level, SBERT is the most lenient of the metrics reported here: two documents that cover the same topics with different wording typically obtain values close to 1.

For *SBERT cosine similarity (ModernBERT)*, Chat yields the highest mean (0.990), with Form and Ask tied at 0.986; aggregated 0.987. The Friedman omnibus is significant ($\chi^2(2) = 16.69, p < 0.001$); Chat > Form and Chat > Ask are both significant with large effects.

In summary, on the document level the three modes split into the patterns *Form* $\approx$ *Chat* > *Ask* (BLEU, ROUGE-1, ROUGE-L, METEOR, BERTScore Precision,

BERTScore F1), *Form* > *Chat* > *Ask* on n-gram overlap of size two and BERTScore Recall, and *Chat* > *Form* $\approx$ *Ask* on SBERT. Ask underperforms both Form and Chat on every metric; Form and Chat trade leadership depending on whether the metric emphasises lexical recall (Form leads) or semantic-document-level fluency (Chat leads). The full within-subjects test results are reproduced in Appendix D.

4.5.2 Chapter-Level Results

The same nine metrics are recomputed at the level of the six Article 30(1) ROPA sections — *Verantwortliche Stelle und Kontaktdaten* (Controller and Contact Details), *Zwecke und Rechtsgrundlagen der Verarbeitung* (Purposes and Legal Bases of Processing), *Kategorien personenbezogener Daten und Datenquellen* (Categories of Personal Data and Sources), *Betroffene Personen und Empfänger* (Data Subjects and Recipients), *Aufbewahrungsfristen und Löschung* (Retention Periods and Deletion), and *Technische und organisatorische Maßnahmen* (Technical and Organisational Measures). Chapter-level results characterise where in the document each mode performs best. Tables 4.10–4.18 report mean (SD) per chapter per mode for each of the nine metrics.

Table 4.10: Chapter-level BLEU mean (SD) by mode (full sample $n = 113$).

Chapter	Form	Ask	Chat	All
Controller & Contact	0.179 (0.077)	0.200 (0.085)	0.166 (0.076)	0.181 (0.080)
Purposes & Legal Bases	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
Personal Data Categories	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
Data Subjects & Recipients	0.013 (0.010)	0.008 (0.014)	0.014 (0.015)	0.012 (0.014)
Retention & Deletion	0.038 (0.044)	0.023 (0.073)	0.068 (0.094)	0.044 (0.076)
Technical & Org. Measures	0.078 (0.068)	0.070 (0.080)	0.091 (0.047)	0.080 (0.066)

Patterns across chapters. *Controller and Contact Details* contains the most templated content (company name, address, DPO contact); the per-mode spread is correspondingly small on every metric, and the absolute scores are the highest of the six chapters on the lexical metrics (BLEU $\approx$ 0.18, ROUGE-1 $\approx$ 0.57, METEOR $\approx$ 0.47). *Purposes and Legal Bases* is the weakest chapter across every metric,

4 Evaluation and Results

Table 4.11: Chapter-level ROUGE-1 mean (SD) by mode (full sample $n = 113$).

Chapter	Form	Ask	Chat	All
Controller & Contact	0.550 (0.097)	0.605 (0.129)	0.559 (0.109)	0.571 (0.114)
Purposes & Legal Bases	0.029 (0.014)	0.031 (0.024)	0.024 (0.012)	0.028 (0.017)
Personal Data Categories	0.056 (0.035)	0.082 (0.062)	0.071 (0.045)	0.070 (0.049)
Data Subjects & Recipients	0.090 (0.034)	0.093 (0.060)	0.123 (0.052)	0.103 (0.052)
Retention & Deletion	0.252 (0.092)	0.212 (0.132)	0.314 (0.143)	0.260 (0.131)
Technical & Org. Measures	0.313 (0.137)	0.330 (0.157)	0.350 (0.090)	0.331 (0.130)

Table 4.12: Chapter-level ROUGE-2 mean (SD) by mode (full sample $n = 113$).

Chapter	Form	Ask	Chat	All
Controller & Contact	0.363 (0.131)	0.430 (0.158)	0.365 (0.138)	0.386 (0.145)
Purposes & Legal Bases	0.014 (0.013)	0.012 (0.016)	0.008 (0.009)	0.011 (0.013)
Personal Data Categories	0.035 (0.030)	0.053 (0.046)	0.036 (0.029)	0.041 (0.036)
Data Subjects & Recipients	0.072 (0.030)	0.059 (0.056)	0.089 (0.049)	0.073 (0.048)
Retention & Deletion	0.134 (0.071)	0.087 (0.107)	0.168 (0.137)	0.131 (0.113)
Technical & Org. Measures	0.166 (0.120)	0.163 (0.141)	0.197 (0.089)	0.176 (0.118)

Table 4.13: Chapter-level ROUGE-L mean (SD) by mode (full sample $n = 113$).

Chapter	Form	Ask	Chat	All
Controller & Contact	0.518 (0.110)	0.576 (0.140)	0.511 (0.128)	0.534 (0.129)
Purposes & Legal Bases	0.028 (0.015)	0.030 (0.022)	0.024 (0.012)	0.027 (0.017)
Personal Data Categories	0.056 (0.035)	0.082 (0.062)	0.071 (0.045)	0.070 (0.049)
Data Subjects & Recipients	0.090 (0.034)	0.093 (0.060)	0.123 (0.052)	0.103 (0.052)
Retention & Deletion	0.218 (0.087)	0.183 (0.141)	0.283 (0.152)	0.229 (0.136)
Technical & Org. Measures	0.237 (0.140)	0.246 (0.158)	0.285 (0.100)	0.257 (0.135)

Table 4.14: Chapter-level METEOR mean (SD) by mode (full sample $n = 113$).

Chapter	Form	Ask	Chat	All
Controller & Contact	0.456 (0.120)	0.506 (0.136)	0.453 (0.124)	0.471 (0.128)
Purposes & Legal Bases	0.034 (0.037)	0.034 (0.047)	0.015 (0.019)	0.027 (0.037)
Personal Data Categories	0.079 (0.073)	0.140 (0.116)	0.101 (0.086)	0.106 (0.096)
Data Subjects & Recipients	0.199 (0.078)	0.203 (0.134)	0.270 (0.105)	0.225 (0.112)
Retention & Deletion	0.332 (0.114)	0.245 (0.160)	0.362 (0.177)	0.314 (0.160)
Technical & Org. Measures	0.295 (0.108)	0.280 (0.145)	0.305 (0.079)	0.293 (0.113)

Table 4.15: Chapter-level BERTScore Precision mean (SD) by mode (full sample $n = 113$).

Chapter	Form	Ask	Chat	All
Controller & Contact	0.855 (0.015)	0.862 (0.020)	0.855 (0.016)	0.857 (0.017)
Purposes & Legal Bases	0.560 (0.009)	0.570 (0.025)	0.568 (0.016)	0.566 (0.019)
Personal Data Categories	0.670 (0.022)	0.690 (0.040)	0.698 (0.032)	0.686 (0.034)
Data Subjects & Recipients	0.686 (0.018)	0.700 (0.035)	0.712 (0.030)	0.700 (0.030)
Retention & Deletion	0.814 (0.033)	0.817 (0.050)	0.840 (0.037)	0.824 (0.042)
Technical & Org. Measures	0.841 (0.038)	0.858 (0.031)	0.863 (0.027)	0.854 (0.033)

Table 4.16: Chapter-level BERTScore Recall mean (SD) by mode (full sample $n = 113$).

Chapter	Form	Ask	Chat	All
Controller & Contact	0.879 (0.019)	0.887 (0.025)	0.879 (0.022)	0.882 (0.022)
Purposes & Legal Bases	0.670 (0.016)	0.667 (0.017)	0.662 (0.016)	0.666 (0.016)
Personal Data Categories	0.850 (0.012)	0.849 (0.015)	0.847 (0.011)	0.849 (0.013)
Data Subjects & Recipients	0.851 (0.016)	0.838 (0.017)	0.836 (0.022)	0.842 (0.020)
Retention & Deletion	0.890 (0.018)	0.879 (0.023)	0.897 (0.027)	0.889 (0.024)
Technical & Org. Measures	0.889 (0.018)	0.882 (0.032)	0.893 (0.022)	0.888 (0.025)

Table 4.17: Chapter-level BERTScore F1 mean (SD) by mode (full sample $n = 113$).

Chapter	Form	Ask	Chat	All
Controller & Contact	0.867 (0.016)	0.875 (0.022)	0.867 (0.017)	0.869 (0.019)
Purposes & Legal Bases	0.610 (0.009)	0.614 (0.014)	0.611 (0.010)	0.612 (0.012)
Personal Data Categories	0.749 (0.016)	0.761 (0.028)	0.765 (0.022)	0.759 (0.023)
Data Subjects & Recipients	0.760 (0.014)	0.762 (0.024)	0.769 (0.022)	0.764 (0.020)
Retention & Deletion	0.850 (0.024)	0.847 (0.035)	0.867 (0.031)	0.855 (0.031)
Technical & Org. Measures	0.864 (0.027)	0.870 (0.027)	0.878 (0.019)	0.871 (0.025)

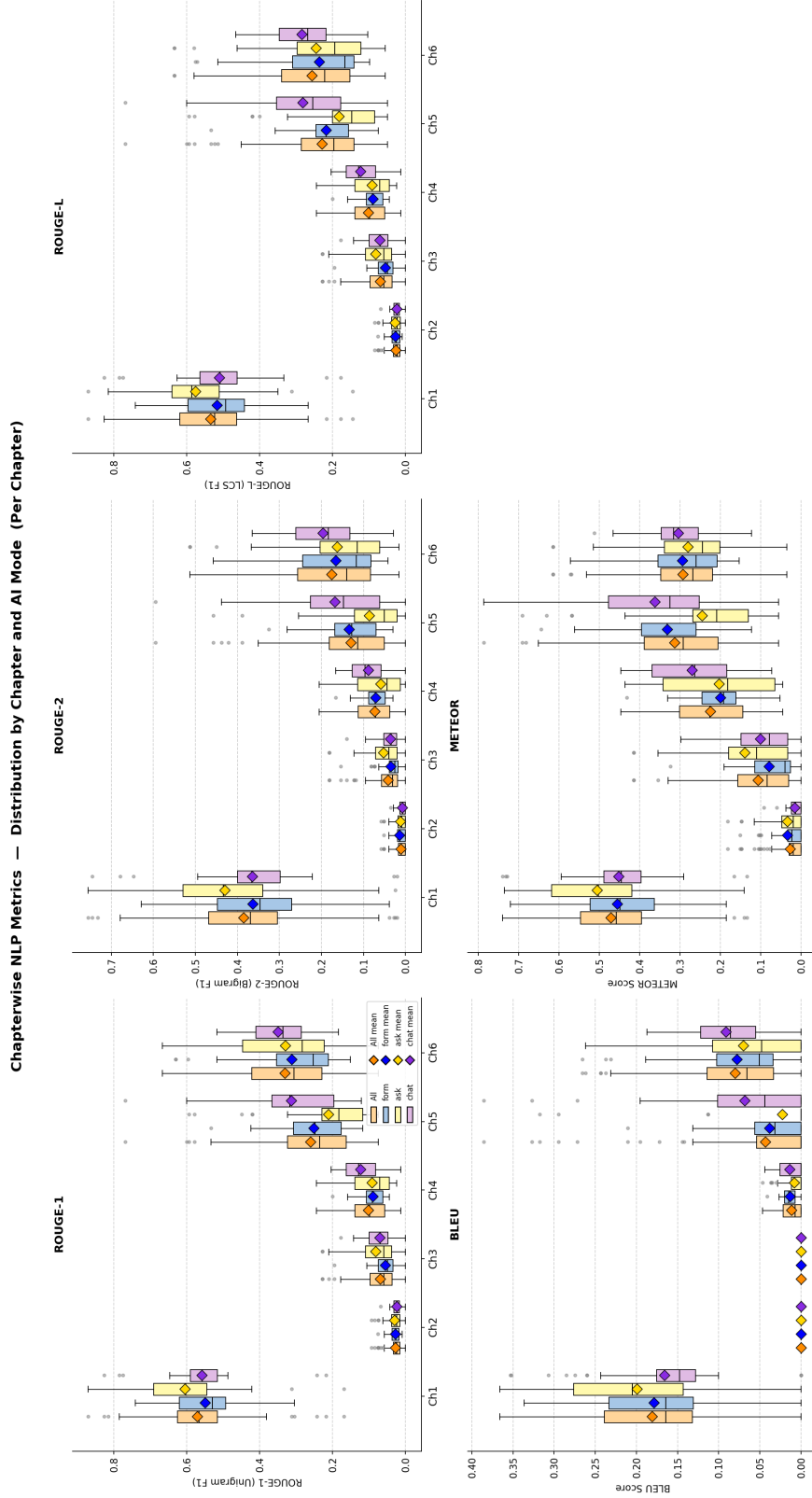


Figure 4.7: Chapter-level distributions of BLEU, ROUGE-1/2/L, and METEOR by mode (full sample $n = 113$).

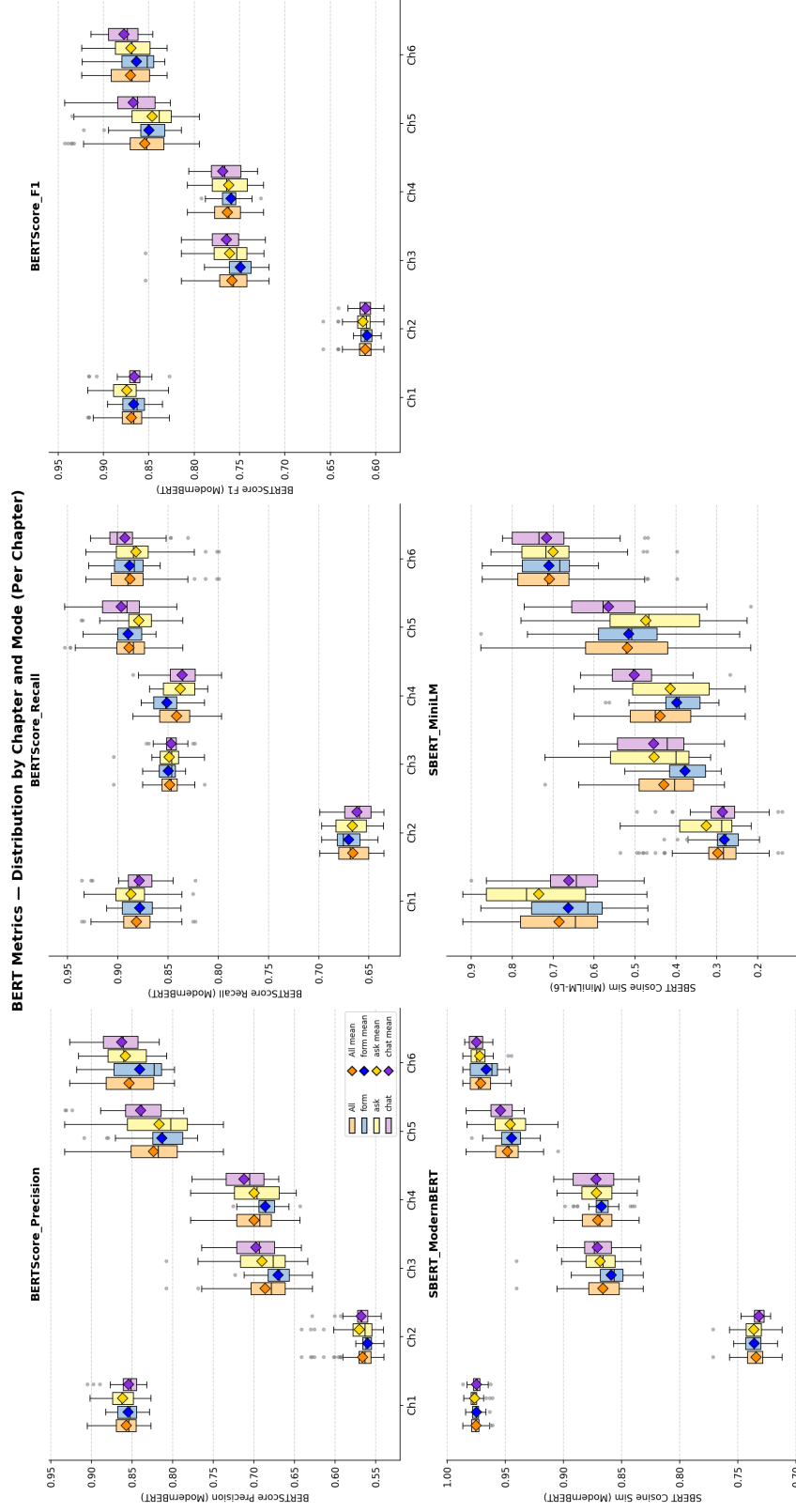


Figure 4.8: Chapter-level distributions of BERTScore Precision / Recall / F1 and SBERT cosine similarity (ModernBERT) by mode (full sample $n = 113$).

Table 4.18: Chapter-level SBERT (ModernBERT) cosine similarity mean (SD) by mode (full sample $n = 113$).

Chapter	Form	Ask	Chat	All
Controller & Contact	0.975 (0.004)	0.977 (0.006)	0.975 (0.005)	0.976 (0.005)
Purposes & Legal Bases	0.736 (0.009)	0.736 (0.012)	0.732 (0.007)	0.735 (0.010)
Personal Data Categories	0.859 (0.015)	0.869 (0.021)	0.871 (0.018)	0.866 (0.019)
Data Subjects & Recipients	0.868 (0.013)	0.872 (0.019)	0.872 (0.022)	0.871 (0.018)
Retention & Deletion	0.945 (0.013)	0.946 (0.018)	0.955 (0.014)	0.949 (0.016)
Technical & Org. Measures	0.967 (0.012)	0.972 (0.010)	0.975 (0.007)	0.971 (0.010)

with BLEU collapsing to zero and METEOR below 0.04 in every mode; this chapter is dominated by free-text legal formulation, and the reference document and participant documents share almost no surface n-grams even where the underlying content overlaps. *Categories of Personal Data* and *Data Subjects and Recipients* sit in an intermediate range (0.06–0.27 on the lexical metrics) with Chat scoring highest on *Data Subjects* (METEOR 0.270 vs 0.199/0.203 for Form/Ask). *Retention Periods* and *Technical and Organisational Measures* are the chapters where Chat establishes its clearest advantage: on Retention, Chat leads on every metric (BERTScore F1 0.867, BLEU 0.068, METEOR 0.362); on T&O Measures, Chat leads on BLEU (0.091), ROUGE-1 (0.350), BERTScore F1 (0.878), and SBERT (0.975).

4.5.3 Document-Level versus Chapter-Level Comparison

A direct comparison of the document-level and chapter-level means is provided by Table 4.19. Document-level means are substantially higher than the cross-chapter means for every metric: for example, the aggregated document-level BERTScore F1 of 0.902 exceeds the average chapter-level F1 of 0.788 by approximately 0.114. The same gap is observable for the lexical metrics, and most pronounced for ROUGE-1 ($\Delta \approx 0.26$). This is consistent with the document-level calculation pooling token overlap across all six chapters, which dilutes chapter-specific zero-overlap regions (notably the Purposes and Legal Bases chapter) and pushes the aggregate scores upward.

Table 4.19: Document-level versus mean chapter-level metric values by mode. The mean chapter-level value is the un-weighted average across the six chapters. Differences are document-level minus chapter-level mean. Source: `doc_vs_chapter_diff.csv`.

Metric	Form		Ask		Chat	
	Doc	Chapter	Doc	Chapter	Doc	Chapter
BLEU	0.129	0.051	0.127	0.050	0.143	0.057
ROUGE-1	0.488	0.215	0.465	0.226	0.511	0.240
ROUGE-2	0.306	0.131	0.257	0.134	0.291	0.144
ROUGE-L	0.388	0.191	0.347	0.202	0.389	0.216
METEOR	0.412	0.232	0.365	0.235	0.390	0.251
BERTScore Precision	0.889	0.738	0.886	0.750	0.895	0.756
BERTScore Recall	0.923	0.838	0.908	0.834	0.913	0.836
BERTScore F1	0.905	0.783	0.897	0.788	0.904	0.793
SBERT (ModernBERT)	0.986	0.892	0.986	0.895	0.990	0.897

4.6 Document Quality — LLM-as-a-Judge

The LLM-as-a-Judge evaluation provides a reference-free, GDPR-specific quality signal across five dimensions: *Completeness*, *Scenario Faithfulness*, *Legal Correctness*, *Hallucination* (inverse-coded; higher equals less hallucination), and *Overall*. The judge is a single LLM (Gemini 3.1 Pro Preview, temperature 0) prompted with the rubric documented in Appendix B.1. All 113 retained documents are scored. Per-mode results are reported in the order Form, Ask, Chat, followed by the aggregated mean across all documents.

4.6.1 Per-Dimension Results

Table 4.20 reports the per-dimension mean and standard deviation per mode and the aggregated mean. Figure 4.9 shows the corresponding distributions.

Table 4.20: LLM-as-Judge dimension means $\pm$ standard deviation by mode (full sample $n = 113$, scale 1–5, higher is better including for *Hallucination (inverse)*). The aggregated “All” column is the mean across all 113 documents.

Dimension	Form	Ask	Chat	All
Completeness	4.84 ± 0.55	3.84 ± 1.54	3.18 ± 1.59	3.94 ± 1.43
Scenario Faithfulness	1.84 ± 0.76	2.30 ± 1.20	2.49 ± 1.19	2.21 ± 1.10
Legal Correctness	3.59 ± 0.55	3.43 ± 1.34	3.15 ± 1.42	3.39 ± 1.18
Hallucination (inverse)	1.65 ± 0.72	2.40 ± 1.12	2.54 ± 0.91	2.20 ± 1.02
Overall	2.41 ± 0.69	2.49 ± 1.04	2.62 ± 1.31	2.50 ± 1.04

For *Completeness*, Form scores highest by a substantial margin (mean 4.84, median 5.0), with Ask second (3.84, median 5.0) and Chat lowest (3.18, median 3.0); aggregated mean 3.94. Completeness decreases by approximately 1.66 points from Form to Chat. Form’s standard deviation is also the lowest of the three modes (0.55 versus Ask 1.54 and Chat 1.59), indicating that Form yields uniformly complete documents.

For *Scenario Faithfulness*, the ordering inverts. Form scores lowest (1.84, median 2.0), Ask second (2.30, median 2.0), Chat highest (2.49, median 2.0); aggregated

4 Evaluation and Results

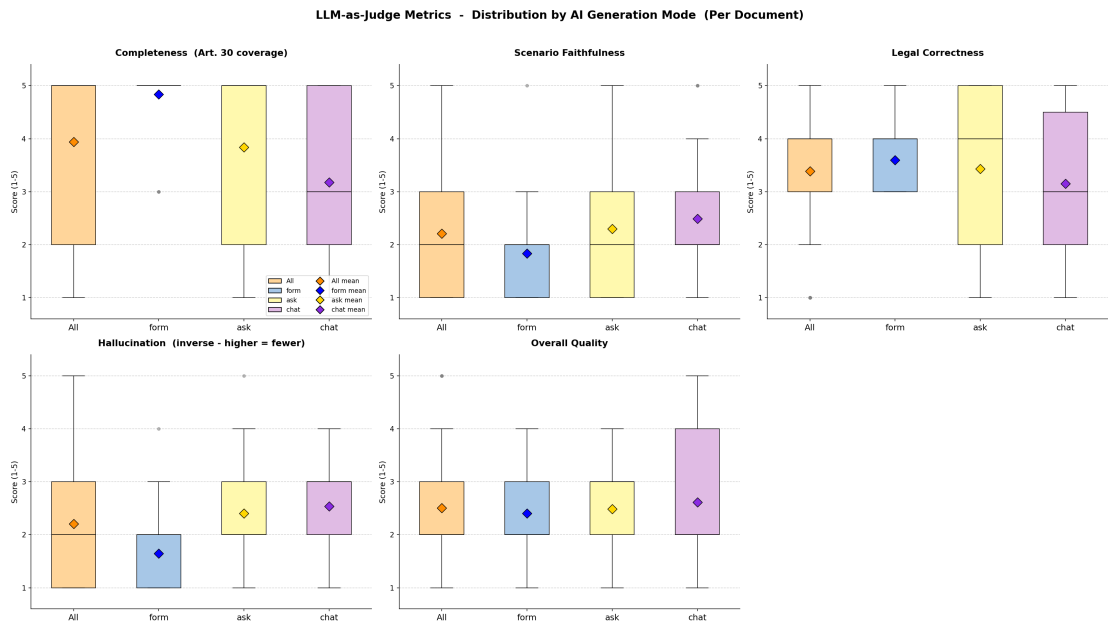


Figure 4.9: Distributions of the five LLM-as-Judge dimensions by mode (full sample $n = 113$, scale 1–5).

mean 2.21. Faithfulness to the specific study scenario increases by approximately 0.65 points from Form to Chat.

For *Legal Correctness*, Form scores highest (3.59, median 4.0), Ask second (3.43, median 4.0), Chat lowest (3.15, median 3.0); aggregated mean 3.39. The spread is approximately 0.45 points across modes; Form’s standard deviation (0.55) is again the lowest, reflecting a narrow concentration of Legal Correctness scores around the value 4.0.

For *Hallucination (inverse)*, the ordering matches Scenario Faithfulness. Form scores lowest (1.65, median 2.0), Ask second (2.40, median 2.0), Chat highest (2.54, median 3.0); aggregated mean 2.20. Because the dimension is inverse-coded, this means Form documents are judged to contain the most fabricated content and Chat documents the least.

For *Overall*, the three modes are clustered closely together: Form 2.41, Ask 2.49, Chat 2.62; aggregated 2.50. The aggregate Overall score thus sits below the mid-point of the 1–5 scale across all modes. Form’s standard deviation (0.69) is again the lowest, while Chat’s is the highest (1.31), indicating that Chat produces both the highest- and lowest-rated documents in the sample.

4.6.2 Cross-Dimension Patterns

Three patterns emerge from the per-dimension means. First, the Completeness ordering (Form > Ask > Chat) is the inverse of the Scenario Faithfulness and Hallucination orderings (Form < Ask < Chat). Form documents are judged to be the most structurally complete but also the most prone to fabricated content, while Chat documents are judged to be the most faithful to the specific scenario but the least structurally complete. Ask sits between the two extremes on every dimension.

Second, the Legal Correctness scores are flat relative to the Completeness spread: the gap between modes is approximately 0.45 points on Legal Correctness compared to 1.66 points on Completeness. The judge therefore distinguishes between modes more sharply on whether the document covers all required Article 30(1) fields than on whether the legal terminology is appropriate.

Third, the Overall scores compress the per-dimension spread into a narrow range (Form 2.41 to Chat 2.62, $\Delta = 0.21$), with the ordering Form < Ask < Chat. This is consistent with Overall reflecting a holistic judgement that weighs Faithfulness and Hallucination against Completeness; Form's high Completeness is partially offset by its low Faithfulness and high Hallucination, and the net result is a slight Chat advantage.

4.6.3 Note on the Form Completeness Pattern

A descriptive observation that recurs across the judge data is that 31 of the 37 Form documents (83.8%) achieve Completeness = 5 while simultaneously scoring Hallucination (inverse) ≤ 2 , indicating “moderate” or “significant” hallucination per the judge rubric. Of these, 17 Form documents combine the maximum Completeness score of 5 with the minimum Hallucination (inverse) score of 1. By contrast, the Ask and Chat distributions on Completeness are more dispersed (standard deviations of 1.54 and 1.59 respectively versus Form's 0.55), and the joint incidence of high Completeness with low Hallucination (inverse) is correspondingly less frequent. This descriptive co-occurrence is the proximate source of the Form-versus-Faithfulness inversion noted above; its interpretation in terms of the checkbox interface design is reserved for the Discussion chapter.

4.7 Summary of Findings

The empirical findings can be summarised as follows. On the user-experience instruments (RQ1), Form is rated more positively than Ask and Chat on the SUS items (notably items 3 “easy to use” and 7 “learn quickly”), reports lower workload than the LLM-driven modes on the R-TLX subscales (especially Temporal Demand and Frustration), and receives the largest share of first-place preference votes (53.6 %). The Friedman omnibus tests on the SUS and R-TLX measures do not reach the conventional significance threshold; the significant sub-effects on the matched-pairs subset are confined to Form versus the LLM-driven modes on Temporal Demand, Frustration, and re-use intent. Ask and Chat are descriptively and inferentially indistinguishable from each other on all aggregate user-experience measures.

On the Perceived AI Support items, the legal-basis confidence item (5_{ai}) increases monotonically with LLM involvement (Form 3.58, Ask 3.74, Chat 3.81; aggregated 3.71), while Form simultaneously yields the highest re-use intent (6_{ai}) at 3.99 versus Ask 3.55 and Chat 3.42; the six AI items pooled together are essentially flat across modes. Free-text comments characterise Form as fast and ergonomic, Ask as transparent but cumbersome, and Chat as comfortable but opaque.

On the document-quality side (RQ2), the reference-based NLP metrics place Form first on lexical recall (BERTScore Recall 0.923 versus Ask 0.908 and Chat 0.913) and on the n-gram-based metrics, Chat first on semantic similarity (SBERT 0.990 versus 0.986 for both other modes) and BLEU, and Ask consistently last on every metric. The chapter-level breakdown attributes Chat’s BLEU and METEOR advantage primarily to the Retention Periods and the Technical and Organisational Measures chapters. The LLM-as-Judge inverts the BERTScore Recall ordering: Form leads on Completeness (4.84) but lags on Scenario Faithfulness (1.84) and Hallucination (1.65, inverse-coded), while Chat leads on the latter two dimensions but trails on Completeness; Overall scores are clustered tightly between 2.41 and 2.62 on the 1–5 scale.

Taken together, the descriptive findings show that on the self-report instruments (RQ1) the structured Form mode is rated most usable, least demanding, and most preferred, while the two LLM-driven modes (Ask and Chat) are descriptively indistinguishable from each other on every aggregate user-experience measure. On the document-quality side (RQ2) the LLM-driven Chat mode achieves the highest

semantic similarity to the reference and the highest Faithfulness and (inverse) Hallucination ratings under the LLM-as-Judge, while the structured Form mode achieves the highest lexical recall and the highest Completeness rating. The interpretation of these patterns against the research questions — in particular the apparent gap between perceived AI support and objective document quality — is taken up in Chapter 5.

4.8 Participant Demographics

This section describes the demographic profile of the matched within-subjects sample of $n = 69$ valid participants ($p_{0001} \geq 1$). Four screened records were excluded from all subsequent analyses for incomplete or invalid participation, yielding the analysis sample reported in Chapter 3. The profile is presented descriptively here; its implications for the generalisability of the findings are reserved for the Discussion chapter.

4.8.1 Numeric and Ordinal Variables

Table 4.21 reports the descriptive statistics for the four numeric or ordinal variables collected in the demographics block: total session duration in seconds, and three five-point Likert self-reports of prior expertise (GDPR knowledge, ROPA familiarity, chatbot familiarity).

The median session duration is 2013 s (≈ 33.6 min) per participant across the full session, consistent with the protocol target of ≈ 90 min total once spread across three modes and the rating instruments. ROPA familiarity is the lowest of the four variables (mean 1.45, median 1), confirming that the sample qualifies as ROPA-novice in line with the inclusion criteria. Chatbot familiarity is correspondingly high (mean 4.36, median 4). GDPR knowledge sits near the scale midpoint (mean 2.81).

Table 4.21: Descriptive statistics for numeric and ordinal demographic variables ($n = 69$). Duration is in seconds and aggregates the full study session across all three modes; the three expertise variables are five-point Likert self-reports (1 = no familiarity, 5 = expert). The duration minimum of -1 is a sentinel value for one non-responding record on this single variable; the median is a more robust point estimate.

Variable	n	Mean	SD	Min	Q1	Median	Q3	Max
Duration (s)	69	1865.7	1354.2	-1	840	2013	2708	5788
GDPR knowledge	69	2.81	1.15	1	2	3	4	5
ROPA familiarity	69	1.45	0.76	1	1	1	2	4
Chatbot familiarity	69	4.36	0.73	2	4	4	5	5

4.8.2 Categorical Variables

Table 4.22 reports counts and shares for the five categorical demographic variables: gender, age (binned by year), highest completed degree, current professional status, and field of study or work. Three participants did not report gender; the share columns therefore use the per-variable response base.

The sample is dominated by Bachelor-level students of IT or Computer Science. Of the 66 participants who reported gender, 62.1% identified as male. The full distribution of demographics and expertise variables is visualised in Figure 4.10.

Table 4.22: Categorical demographic variables ($n = 69$ unless noted). Shares are computed within variable.

Variable	Level	n	Share
Gender ($n = 66$ reporters)	Male	41	62.1 %
	Female	24	36.4 %
	Other	1	1.5 %
Age (years)	24	18	26.1 %
	25	18	26.1 %
	26	10	14.5 %
	Other (21–40)	23	33.3 %
Degree	Bachelor	60	88.2 %
	Master	6	8.8 %
	Other	2	2.9 %
Profession	Student	56	81.2 %
	Employed	10	14.5 %
	PhD	2	2.9 %
	Staatsexamen	1	1.5 %
Field	IT	41	59.4 %
	Business (BWL)	16	23.2 %
	Other	6	8.7 %
	Research	4	5.8 %
	HR	2	2.9 %

4 Evaluation and Results



5 Discussion

5.1 User Study Discussion

User Study Discussion

5.2 Document Discussion

Discussion on quality of generated document

6 Conclusion

A Study Scenario

The scenario reproduced below was presented to all participants in German, regardless of interaction mode (Form, Ask, Chat). The English translation is provided for the international reader; the German original is the version actually shown.

A.1 Original (German)

Szenario: Mitarbeiterverwaltung (Arbeitszeit/Urlaub) & Parkplatzberechtigung

Sie sind Mitarbeitender namens A.I. und arbeiten für die APOR GmbH, ein Softwareunternehmen mit ca. 50 Mitarbeitenden in Ulm. Sie handeln im Namen des Unternehmens und sind im Szenario sowohl Verantwortlicher (Data Controller) als auch Datenschutzbeauftragte:r (DPO/DSB).

Zur Organisation:

Rolle: Data Controller, DPO

Adresse: Mainstraße 67, Ulm

E-Mail: mail@apor.eu

Telefon: 012345678

Vertreter: (Sie), A. I., ai@apor.eu

DPO: Das sind auch Sie

Verarbeitung

Das Unternehmen verarbeitet personenbezogene Daten, um Mitarbeitende vom Eintritt bis zum Austritt zu verwalten. Im Fokus stehen:

- Anlegen neuer Mitarbeitender
- Dokumentation von Arbeitszeiten
- Verwaltung von Urlaub und Abwesenheiten

Zusätzlich verwaltet APOR für berechnigte Mitarbeitende eine Parkplatzberechnigung für einen Firmenparkplatz mit Schranke. Damit berechnigte Mitarbeitende den Parkplatz nutzen können, wird ein geeignetes Merkmal zur Zuordnung der Berechnigung hinterlegt (z. B. ein Fahrzeugkennzeichen).

APOR arbeitet mit intern gehostetem Microsoft Office sowie internen Servern (keine externe Cloud). Krankentage oder medizinische Angaben sind für diese Verarbeitung nicht erforderlich.

Für die Verwaltung sind technische und organisatorische Maßnahmen vorgesehen, wie z.B. Zugriffsbeschränkung nach Rollen (Need-to-know), Passwortschutz (ggf. MFA), regelmäßige Backups, Zutrittskontrolle zu Serverraum/Archiv und Vertraulichkeitsverpflichtung der berechtigten Mitarbeitenden.

Alle in diesem Zusammenhang gespeicherten Informationen werden spätestens 3 Monate nach Austritt der jeweiligen Person gelöscht.

A.2 English Translation

Scenario: Employee Administration (Working Time / Leave) & Parking Authorisation

You are an employee named A.I., working for APOR GmbH, a software company with approximately 50 employees in Ulm. You act on behalf of the company and, in this scenario, are both the Data Controller and the Data Protection Officer (DPO).

Organisation details:

Role: Data Controller, DPO

Address: Mainstrasse 67, Ulm

E-mail: mail@apor.eu

Phone: 012345678

Representative: (You), A. I., ai@apor.eu

DPO: Also you

Processing

The company processes personal data in order to administer employees from onboarding to exit. The focus is on:

- Onboarding new employees
- Documentation of working hours
- Administration of leave and absences

In addition, APOR manages a parking authorisation for entitled employees for a company car park with a barrier. To allow entitled employees to use the car park, a suitable feature for assigning the authorisation is recorded (e.g. a vehicle licence plate).

APOR uses internally hosted Microsoft Office and internal servers (no external cloud). Sick days or medical information are not required for this processing.

For administration, technical and organisational measures (TOMs) are provided, such as role-based access restriction (need-to-know), password protection (where appropriate, MFA), regular backups, physical access control to the server room / archive, and a confidentiality obligation for authorised employees.

All information stored in this context is deleted no later than 3 months after the person's departure.

B LLM-as-a-Judge Configuration

This appendix documents the inputs to the LLM-as-a-judge evaluation layer that scored the 113 retained ROPA documents. It contains the verbatim system and user prompts, the five-dimension rubric, and the model configuration needed to reproduce the evaluation.

B.1 System and User Prompt

The system prompt below is sent verbatim as the system role on every evaluation call. It is taken from `python/judge/prompt.json` (key `judge.system`).

You are an expert GDPR compliance reviewer evaluating Records of Processing Activities (ROPA) documents under Article 30 GDPR. The documents are written by novices (non-experts, no prior GDPR training) who described the same processing scenario using different LLM-assisted tools. Your job is to judge each ROPA document against (a) the provided scenario and (b) GDPR Article 30 requirements. Be strict but fair. Do not reward verbosity. Do not penalise a document for merely restating scenario facts. Respond with valid JSON only - no prose, no Markdown fences, no commentary before or after the JSON object.

The accompanying user-message template is sent on every call with `{scenario}` (the German original from Appendix A.1) and `{document}` (the participant ROPA document) substituted in:

You will evaluate one ROPA document against the scenario it is supposed

to cover.

```
=== SCENARIO (verbatim, German) ===  
{scenario}  
=== END SCENARIO ===
```

```
=== ROPA DOCUMENT TO EVALUATE ===  
{document}  
=== END DOCUMENT ===
```

Score the document on each of the following five metrics using a 1-5 Likert scale, and give a short rationale (max 2 sentences, English) for each score. The scale for all metrics is:

1 = very poor, 2 = poor, 3 = adequate, 4 = good, 5 = excellent.

Metrics and what to assess:

- completeness: Coverage of Art. 30(1) GDPR required elements - controller identity and contact, DPO, purposes of processing, legal basis, categories of data subjects and personal data, categories of recipients, retention periods, and technical/organisational measures (TOMs). Missing elements lower the score.
- scenario_faithfulness: How accurately the document reflects the concrete facts of THIS scenario - APOR GmbH, Ulm (Mainstrasse 67), ~50 employees, internally hosted Microsoft Office and internal servers (NO external cloud), no medical/sickness data, 3-month deletion after employee departure, parking lot authorisation using a distinguishing feature such as a licence plate. Generic ROPA boilerplate that ignores these specifics lowers the score.
- legal_correctness: Correct use of GDPR terminology and correct citation of legal basis (typically Art. 6(1)(b) performance of contract for employment data, Art. 6(1)(c) legal obligation where applicable, Art. 6(1)(f) legitimate interest for parking authorisation). Wrong articles, invented legal bases, or confusing controller/processor/DPO roles lower the score.
- hallucination_inverse: Inverse score for invented facts not supported

by the scenario (e.g. invented third-party processors, invented data categories like medical data, invented addresses, invented retention periods different from 3 months). 5 = no hallucinations, 1 = many invented facts.

- overall: Holistic judgement of how useful this document would be as a real Art. 30 ROPA record for this scenario, taking the four dimensions above into account.

Return a single JSON object with exactly this shape:

```
{
  "completeness":          {"score": <int 1-5>, "rationale": "<string>"},
  "scenario_faithfulness": {"score": <int 1-5>, "rationale": "<string>"},
  "legal_correctness":     {"score": <int 1-5>, "rationale": "<string>"},
  "hallucination_inverse": {"score": <int 1-5>, "rationale": "<string>"},
  "overall":               {"score": <int 1-5>, "rationale": "<string>"}
}
```

B.2 Rubric — Five Dimensions

All five dimensions share the same 1–5 Likert anchor: 1 = very poor, 2 = poor, 3 = adequate, 4 = good, 5 = excellent. The fourth dimension, `hallucination_inverse`, is inverted internally so that for all five dimensions higher equals better. For each dimension the judge returns both an integer score and a short English rationale (maximum two sentences).

B.3 Model Configuration and Limitations

The configuration in Table B.2 is taken from the `judge` block of `python/judge/prompt.-json` and the corresponding cells of `ROPAGEN_JUDGE.ipynb`. It is the minimal information required to reproduce the evaluation layer.

Dimension	What is assessed
Completeness	Coverage of Art. 30(1) GDPR required elements: controller identity and contact, DPO, purposes of processing, legal basis, categories of data subjects and personal data, categories of recipients, retention periods, and technical/organisational measures (TOMs). Missing elements lower the score.
Scenario Faithfulness	Accuracy with respect to the concrete facts of this scenario: APOR GmbH, Ulm (Mainstraße 67), ~50 employees, internally hosted Microsoft Office and internal servers (no external cloud), no medical/sickness data, 3-month deletion after employee departure, parking authorisation using a distinguishing feature such as a licence plate. Generic ROPA boilerplate lowers the score.
Legal Correctness	Correct use of GDPR terminology and correct citation of legal basis (typically Art. 6(1)(b) for employment data, Art. 6(1)(c) where applicable, Art. 6(1)(f) for parking authorisation). Wrong articles, invented legal bases, or confused controller/processor/DPO roles lower the score.
Hallucination (inverse)	Inverse score for invented facts not supported by the scenario (e.g. invented third-party processors, invented categories such as medical data, invented addresses, retention periods other than 3 months). 5 = no hallucinations, 1 = many invented facts.
Overall	Holistic judgement of how useful the document would be as a real Art. 30 ROPA record for this scenario, taking the four dimensions above into account.

Table B.1: The five judge dimensions. All share the 1–5 Likert anchor; higher is better for every dimension after the hallucination inversion.

B LLM-as-a-Judge Configuration

Setting	Value
Model	google/gemini-3.1-pro-preview (via OpenRouter)
Temperature	0.0 (deterministic decoding)
Output format	JSON object, enforced via OpenRouter <code>response_format = json_schema</code> with <code>strict: true</code>
Schema name	ropa_judge_result
Required keys	completeness, scenario_faithfulness, legal_correctness, hallucination_inverse, overall
Per-key shape	<code>{"score": int in [1,5], "rationale": string}</code>
Additional properties	Disallowed at every object level
Documents evaluated	113 ROPA documents (Form 37, Ask 37, Chat 39)
Caching	Per-document JSON result cached on disk; reused on re-runs
Retry behaviour	Schema-validation or transient API failures retried; persistent failures logged and skipped
Logging	Each call logs model, latency, prompt and completion token counts, and raw response payload

Table B.2: LLM-as-a-judge model configuration.

Known limitations. The following limitations of the judge layer are stated for full transparency and discussed further in the Discussion chapter:

- **Single-judge bias.** Only one judge model (Gemini 3.1 Pro Preview) operationalises the rubric; inter-rater agreement with a human GDPR expert is not established.
- **Model-family bias.** Documents are produced by Mistral Large 3 inside ropagen, while the judge belongs to a different model family (Google). This reduces but does not eliminate stylistic-preference bias.
- **Single-scenario bias.** All scores are conditional on the one APOR GmbH scenario; generalisation to other ROPA contexts is not claimed.
- **Reference-free.** Unlike the NLP metric layer, the judge does not compare against an expert reference document; it scores against the scenario and Art. 30 directly.

C Full Inferential Tables

This appendix consolidates the canonical sample-size derivation and the full inferential statistics for both the survey and the NLP/judge layer. The tables in the Evaluation chapter draw selectively from these; the appendix is reproduced for completeness and reproducibility.

C.1 Sample-Size Derivation

All sample sizes used in the thesis are derived from the raw data files committed to the repository: `python/survey/docs/data.csv` for the survey (one row per participant) and `python/metrics/output/all_scores_combined.csv` for the NLP and judge scores (one row per AI-generated document).

Survey subset	n
Recruited participants (raw)	73
Participants with at least one valid session ($p_{0001} \geq 1$)	70
Form responses (mode-level)	69
Ask responses (mode-level)	69
Chat responses (mode-level)	69
Matched-triple participants (valid response in all three modes)	69

Table C.1: Survey sample sizes after consent and quality filtering.

C.2 Survey Inferential Tables

All survey tests use the matched-triple $n = 69$ within-subjects subset. Omnibus is the Friedman test (non-parametric repeated-measures); post-hoc is pairwise Wilcoxon

NLP / judge subset	n
Total documents with parseable NLP scores and judge ratings	113
Form documents	37
Ask documents	37
Chat documents	39
Unique participants represented	47
Participants with at least one document in each mode	28
Strict matched triple (exactly one document per mode)	26

Table C.2: NLP and judge sample sizes. The strict $n = 26$ set is used for all within-subjects tests (Friedman, Wilcoxon signed-rank); the unbalanced $n = 113$ is used for between-groups sensitivity checks.

Survey $\times$ NLP join	n
Joined rows (document with matching survey response)	106
Form joined rows	34
Ask joined rows	34
Chat joined rows	38
Unique participants in the joined sample	44
Participants with survey $\times$ NLP coverage in all three modes	27

Table C.3: Triangulated samples used for confidence–quality analyses. The joined $n = 106$ is the canonical sample for the rank-inversion table; the within-subjects $n = 27$ is used for paired confidence $\times$ quality correlations.

signed-rank with Bonferroni-corrected $\alpha = 0.0167$. Effect-size labels: $|r| < 0.1$ negligible, 0.1–0.3 small, 0.3–0.5 medium, ≥ 0.5 large. For Kendall's W : 0–0.1 weak, 0.1–0.3 moderate, 0.3–0.5 strong.

Measure	n	χ^2	df	p	W	Sig.
SUS (0–100, higher = better)	69	5.51	2	0.0636	0.040	No
R-TLX composite (1–21)	69	5.24	2	0.0728	0.038	No
R-TLX Mental Demand	69	3.13	2	0.2086	0.023	No
R-TLX Physical Demand	69	2.08	2	0.3535	0.015	No
R-TLX Temporal Demand	69	12.77	2	0.0017	0.093	Yes
R-TLX Effort	69	3.78	2	0.1511	0.027	No
R-TLX Frustration	69	8.93	2	0.0115	0.065	Yes
R-TLX Performance (positive-coded)	69	2.33	2	0.3114	0.017	No
AI Confidence (item 5_ai)	69	3.76	2	0.1524	0.027	No
AI Support (6-item composite)	69	1.05	2	0.5919	0.008	No
AI Reuse intention (item 6_ai)	69	11.12	2	0.0039	0.081	Yes

Table C.4: Friedman omnibus across all survey measures ($n = 69$ matched triple).

Mode preference ranking ($n = 69$). The chi-square goodness-of-fit on first-rank votes against a uniform null gives $\chi^2(2, N=69) = 14.17$, $p < 0.001$, Cramér's $V = 0.320$. The Friedman test on the full rank scores (1–3) gives $\chi^2(2) = 5.07$, $p = 0.0792$, $W = 0.037$. Per-mode rank-1 vote counts: Form 37, Ask 12, Chat 20; mean ranks (lower = better): Form 1.78, Chat 2.07, Ask 2.14.

C.3 NLP and Judge Inferential Tables

NLP and judge tests follow the same family: Friedman omnibus on the strict matched-triple $n = 26$, Wilcoxon signed-rank pairwise post-hoc with Bonferroni-corrected $\alpha = 0.0167$, plus a Kruskal–Wallis sensitivity check on the full unbalanced $n = 113$. Shapiro–Wilk testing identified non-normal distributions in BLEU, BERTScore Recall (Ask, Chat), and SBERT (Form, Ask); non-parametric tests are used throughout.

Measure	Pair	n	W	p	r	Sig. (Bonf.)
SUS	Form vs Ask	60	578.5	0.0131	+0.368	Yes
SUS	Form vs Chat	58	549.5	0.0177	+0.358	No
SUS	Ask vs Chat	61	885.5	0.6660	−0.063	No
R-TLX comp.	Form vs Ask	65	684.0	0.0111	−0.362	Yes
R-TLX comp.	Form vs Chat	65	636.0	0.0043	−0.407	Yes
R-TLX comp.	Ask vs Chat	63	995.5	0.9318	+0.012	No
Temporal Demand	Form vs Ask	43	237.0	0.0038	−0.499	Yes
Temporal Demand	Form vs Chat	46	233.0	<0.001	−0.569	Yes
Temporal Demand	Ask vs Chat	46	427.0	0.2111	−0.210	No
Frustration	Form vs Ask	52	403.5	0.0091	−0.414	Yes
Frustration	Form vs Chat	57	519.5	0.0144	−0.371	Yes
Frustration	Ask vs Chat	52	629.0	0.5838	+0.087	No
AI Reuse	Form vs Ask	49	402.5	0.0339	+0.343	No
AI Reuse	Form vs Chat	43	216.0	0.0015	+0.543	Yes
AI Reuse	Ask vs Chat	45	454.5	0.4684	+0.122	No

Table C.5: Pairwise Wilcoxon signed-rank tests for the survey measures with at least one significant pairwise effect. Sign convention: positive r means the first-named mode scores higher on the measure. Bonferroni-corrected $\alpha = 0.0167$.

Metric	H	p	Sig.	ϵ^2
BLEU	4.48	0.107	No	—
ROUGE-1	5.57	0.062	No	—
ROUGE-2	9.56	0.008	Yes	0.069 (mod.)
ROUGE-L	7.79	0.020	Yes	—
METEOR	9.80	0.007	Yes	0.071 (mod.)
BERTScore Precision	9.11	0.011	Yes	—
BERTScore Recall	32.38	<0.001	Yes	0.276 (large)
BERTScore F1	9.75	0.008	Yes	0.071 (mod.)
SBERT	17.43	<0.001	Yes	0.140 (large)

Table C.6: Kruskal–Wallis sensitivity check on the full unbalanced sample ($n = 113$).

Metric	χ^2	p	Sig.	W
BLEU	6.23	0.044	Yes	0.083 (weak)
ROUGE-1	10.69	0.005	Yes	~0.18 (mod.)
ROUGE-2	13.46	0.001	Yes	~0.22 (mod.)
ROUGE-L	11.31	0.004	Yes	~0.19 (mod.)
METEOR	9.92	0.007	Yes	~0.17 (mod.)
BERTScore Precision	18.54	<0.001	Yes	~0.30 (strong)
BERTScore Recall	30.77	<0.001	Yes	0.410 (strong)
BERTScore F1	13.00	0.002	Yes	~0.21 (mod.)
SBERT	16.69	<0.001	Yes	~0.27 (mod.)

Table C.7: Friedman omnibus on the strict within-subjects sample ($n = 26$).

Confidence–quality correlations. Spearman ρ between item 5_ai (AI legal-basis confidence) and document quality, computed within each mode on the joined survey $\times$ NLP sample. Form ($n = 34$): $\rho = -0.22$ ($p = 0.22$) vs. BERTScore F1; $\rho = +0.07$ ($p = 0.70$) vs. Judge Overall. Ask ($n = 34$): $\rho = -0.23$ ($p = 0.19$) vs. BERTScore F1; $\rho = -0.21$ ($p = 0.24$) vs. Judge Overall. Chat ($n = 38$): $\rho = +0.28$ ($p = 0.09$) vs. BERTScore F1; $\rho = +0.41$ ($p = 0.012$) vs. Judge Overall. Pooled ($n = 106$): $\rho = -0.02$ ($p = 0.88$) vs. BERTScore F1; $\rho = +0.14$ ($p = 0.14$) vs. Judge Overall.

Metric	Pair	W	p	r	Sig. (Bonf.)
BLEU	Form vs Ask	108	0.089	+0.385	No
BLEU	Form vs Chat	133	0.291	−0.242	No
BLEU	Ask vs Chat	74	0.009	−0.578	Yes
ROUGE-1	Form vs Ask	77	0.011	+0.561	Yes
ROUGE-1	Form vs Chat	147	0.483	−0.162	No
ROUGE-1	Ask vs Chat	63	0.003	−0.641	Yes
ROUGE-2	Form vs Ask	30	<0.001	+0.829	Yes
ROUGE-2	Form vs Chat	136	0.328	+0.225	No
ROUGE-2	Ask vs Chat	71	0.007	−0.595	Yes
ROUGE-L	Form vs Ask	44	<0.001	+0.749	Yes
ROUGE-L	Form vs Chat	161	0.727	+0.083	No
ROUGE-L	Ask vs Chat	66	0.004	−0.624	Yes
METEOR	Form vs Ask	33	<0.001	+0.812	Yes
METEOR	Form vs Chat	137	0.340	+0.219	No
METEOR	Ask vs Chat	84	0.019	−0.521	No
BERTScore Precision	Form vs Ask	63	0.003	+0.641	Yes
BERTScore Precision	Form vs Chat	106	0.080	−0.396	No
BERTScore Precision	Ask vs Chat	40	<0.001	−0.772	Yes
BERTScore Recall	Form vs Ask	0	<0.001	+1.000	Yes
BERTScore Recall	Form vs Chat	55	0.001	+0.687	Yes
BERTScore Recall	Ask vs Chat	99	0.053	−0.436	No
BERTScore F1	Form vs Ask	24	<0.001	+0.863	Yes
BERTScore F1	Form vs Chat	172	0.940	+0.020	No
BERTScore F1	Ask vs Chat	59	0.002	−0.664	Yes
SBERT	Form vs Ask	110	0.099	+0.373	No
SBERT	Form vs Chat	56	0.002	−0.681	Yes
SBERT	Ask vs Chat	26	<0.001	−0.852	Yes

Table C.8: Pairwise Wilcoxon signed-rank tests on the strict within-subjects sample ($n = 26$). Sign convention: positive r means the first-named mode scores higher on the metric. Bonferroni-corrected $\alpha = 0.0167$.

D Inferential Statistics in Full

This appendix reproduces the complete inferential test output referenced in Chapter 4. Sample-size definitions follow Chapter 3: the within-subjects survey tests use $n = 69$, the within-subjects document-quality tests use the matched-triple subset of $n = 26$, and between-groups sensitivity checks on the unbalanced full sample use $n = 113$.

D.1 Survey Inferential Tests

D.1.1 Normality (Shapiro–Wilk, $n = 69$)

Table D.1: Shapiro–Wilk normality tests for survey measures by mode.

Measure	Mode	W	p	Normal?
SUS	Form	0.944	0.004	No
SUS	Ask	0.976	0.210	Yes
SUS	Chat	0.927	0.001	No
R-TLX composite	Form	0.940	0.003	No
R-TLX composite	Ask	0.926	0.001	No
R-TLX composite	Chat	0.945	0.005	No
AI Support (6-item)	Form	0.923	< 0.001	No
AI Support (6-item)	Ask	0.965	0.048	No
AI Support (6-item)	Chat	0.952	0.010	No
AI Confidence (5_{ai})	Form	0.846	< 0.001	No
AI Confidence (5_{ai})	Ask	0.856	< 0.001	No
AI Confidence (5_{ai})	Chat	0.864	< 0.001	No
AI Reuse (6_{ai})	Form	0.799	< 0.001	No
AI Reuse (6_{ai})	Ask	0.872	< 0.001	No
AI Reuse (6_{ai})	Chat	0.861	< 0.001	No

Non-normal distributions are detected for nearly every measure. Non-parametric tests (Friedman omnibus and Wilcoxon signed-rank post-hoc) are used throughout.

D.1.2 Friedman Omnibus ($n = 69$ matched triple)

Table D.2: Friedman omnibus tests across the three modes for survey measures. Kendall's W : 0–0.1 weak, 0.1–0.3 moderate, 0.3–0.5 strong.

Measure	χ^2	df	p	W	Significant
SUS (0–100)	5.51	2	0.064	0.040	No
R-TLX composite (1–21)	5.24	2	0.073	0.038	No
R-TLX Mental Demand	3.13	2	0.209	0.023	No
R-TLX Physical Demand	2.08	2	0.354	0.015	No
R-TLX Temporal Demand	12.77	2	0.002	0.093	Yes
R-TLX Effort	3.78	2	0.151	0.027	No
R-TLX Frustration	8.93	2	0.012	0.065	Yes
R-TLX Performance (pos.)	2.33	2	0.311	0.017	No
AI Confidence (5_{ai})	3.76	2	0.152	0.027	No
AI Support (6-item composite)	1.05	2	0.592	0.008	No
AI Reuse intention (6_{ai})	11.12	2	0.004	0.081	Yes

D.1.3 Pairwise Wilcoxon Signed-Rank Post-hoc (Bonferroni

$$\alpha = 0.0167)$$

Post-hoc tests are reported for measures with a significant Friedman result and, for completeness, for SUS and the R-TLX composite. Sign convention: positive r means the first-named mode ranks higher. Effect-size labels: $|r| < 0.1$ negligible, 0.1–0.3 small, 0.3–0.5 medium, ≥ 0.5 large.

D.1.4 Mode Preference Ranking

A chi-square goodness-of-fit test on the rank-1 vote distribution against a uniform null gives $\chi^2(2, N = 69) = 14.17$, $p < 0.001$, Cramér's $V = 0.320$ (medium effect). The Friedman test on the full rank scores (1–3) is non-significant: $\chi^2(2) = 5.07$, $p = 0.079$, $W = 0.037$.

Table D.3: Pairwise Wilcoxon signed-rank post-hoc tests for survey measures.

Measure	Pair	n	W	p	r	Sig (Bonf)
SUS	Form vs Ask	60	578.5	0.013	+0.368 medium	Yes
SUS	Form vs Chat	58	549.5	0.018	+0.358 medium	No
SUS	Ask vs Chat	61	885.5	0.666	−0.063 negligible	No
R-TLX composite	Form vs Ask	65	684.0	0.011	−0.362 medium	Yes
R-TLX composite	Form vs Chat	65	636.0	0.004	−0.407 medium	Yes
R-TLX composite	Ask vs Chat	63	995.5	0.932	+0.012 negligible	No
Temporal Demand	Form vs Ask	43	237.0	0.004	−0.499 medium	Yes
Temporal Demand	Form vs Chat	46	233.0	< 0.001	−0.569 large	Yes
Temporal Demand	Ask vs Chat	46	427.0	0.211	−0.210 small	No
Frustration	Form vs Ask	52	403.5	0.009	−0.414 medium	Yes
Frustration	Form vs Chat	57	519.5	0.014	−0.371 medium	Yes
Frustration	Ask vs Chat	52	629.0	0.584	+0.087 negligible	No
AI Reuse (6_{ai})	Form vs Ask	49	402.5	0.034	+0.343 medium	No
AI Reuse (6_{ai})	Form vs Chat	43	216.0	0.002	+0.543 large	Yes
AI Reuse (6_{ai})	Ask vs Chat	45	454.5	0.468	+0.122 small	No

D.2 NLP Metric Inferential Tests

D.2.1 Kruskal–Wallis Sensitivity Check ($n = 113$, unbalanced)

Table D.4: Kruskal–Wallis H tests on the unbalanced full document set. Reported as a between-groups sensitivity check, not the primary post-hoc family.

Metric	H	p	Sig ($\alpha = 0.05$)	ε^2
BLEU	4.48	0.107	No	—
ROUGE-1	5.57	0.062	No	—
ROUGE-2	9.56	0.008	Yes	0.069 (moderate)
ROUGE-L	7.79	0.020	Yes	—
METEOR	9.80	0.007	Yes	0.071 (moderate)
BERTScore Precision	9.11	0.011	Yes	—
BERTScore Recall	32.38	< 0.001	Yes	0.276 (large)
BERTScore F1	9.75	0.008	Yes	0.071 (moderate)
SBERT (ModernBERT)	17.43	< 0.001	Yes	0.140 (large)

Table D.5: Friedman omnibus tests on the matched-triple subset of $n = 26$ participants. Kendall's W approximate where the metric required tie correction.

Metric	χ^2	p	Sig	W
BLEU	6.23	0.044	Yes	0.083 (weak)
ROUGE-1	10.69	0.005	Yes	~ 0.18 (moderate)
ROUGE-2	13.46	0.001	Yes	~ 0.22 (moderate)
ROUGE-L	11.31	0.004	Yes	~ 0.19 (moderate)
METEOR	9.92	0.007	Yes	~ 0.17 (moderate)
BERTScore Precision	18.54	< 0.001	Yes	~ 0.30 (strong)
BERTScore Recall	30.77	< 0.001	Yes	0.410 (strong)
BERTScore F1	13.00	0.002	Yes	~ 0.21 (moderate)
SBERT (ModernBERT)	16.69	< 0.001	Yes	~ 0.27 (moderate)

D.2.2 Friedman Omnibus ($n = 26$ matched triple)

D.2.3 Pairwise Wilcoxon Signed-Rank Post-hoc ($n = 26$, primary)

Bonferroni-corrected $\alpha = 0.0167$. Effect size: matched-pairs rank-biserial r (positive when the first-named mode in the pair scores higher).

D.2.4 Mann–Whitney U Sensitivity Check ($n = 113$, unbalanced)

The Mann–Whitney U test is reported as a between-groups sensitivity check on the unbalanced full sample. Bonferroni-corrected $\alpha = 0.0167$. Only significant comparisons are listed; non-significant comparisons are omitted for brevity.

The sensitivity check broadly agrees with the matched-pairs result on direction (Ask $<$ Form, Ask $<$ Chat on most metrics), with the matched analysis producing larger and more consistent effect sizes — as expected for a within-subjects design.

D.3 Confidence–Quality Correlations

Spearman rank correlation ρ between the primary confidence measure (item 5_{ai} , “I am confident the AI identified the correct legal basis”) and document-quality mea-

Table D.6: Pairwise Wilcoxon signed-rank post-hoc tests for document-level NLP metrics on the matched-triple subset.

Metric	Pair	W	p	r	Direction	Sig (Bonf)
BLEU	Form vs Ask	108	0.089	+0.385 medium	Form > Ask	No
BLEU	Form vs Chat	133	0.291	−0.242 small	Chat > Form	No
BLEU	Ask vs Chat	74	0.009	−0.578 large	Chat > Ask	Yes
ROUGE-1	Form vs Ask	77	0.011	+0.561 large	Form > Ask	Yes
ROUGE-1	Form vs Chat	147	0.483	−0.162 small	Chat > Form	No
ROUGE-1	Ask vs Chat	63	0.003	−0.641 large	Chat > Ask	Yes
ROUGE-2	Form vs Ask	30	< 0.001	+0.829 large	Form > Ask	Yes
ROUGE-2	Form vs Chat	136	0.328	+0.225 small	Form > Chat	No
ROUGE-2	Ask vs Chat	71	0.007	−0.595 large	Chat > Ask	Yes
ROUGE-L	Form vs Ask	44	< 0.001	+0.749 large	Form > Ask	Yes
ROUGE-L	Form vs Chat	161	0.727	+0.083 negligible	Form > Chat	No
ROUGE-L	Ask vs Chat	66	0.004	−0.624 large	Chat > Ask	Yes
METEOR	Form vs Ask	33	< 0.001	+0.812 large	Form > Ask	Yes
METEOR	Form vs Chat	137	0.340	+0.219 small	Form > Chat	No
METEOR	Ask vs Chat	84	0.019	−0.521 large	Chat > Ask	No (above B
BERTScore Precision	Form vs Ask	63	0.003	+0.641 large	Form > Ask	Yes
BERTScore Precision	Form vs Chat	106	0.080	−0.396 medium	Chat > Form	No
BERTScore Precision	Ask vs Chat	40	< 0.001	−0.772 large	Chat > Ask	Yes
BERTScore Recall	Form vs Ask	0	< 0.001	+1.000 large	Form > Ask	Yes
BERTScore Recall	Form vs Chat	55	0.001	+0.687 large	Form > Chat	Yes
BERTScore Recall	Ask vs Chat	99	0.053	−0.436 medium	Chat > Ask	No
BERTScore F1	Form vs Ask	24	< 0.001	+0.863 large	Form > Ask	Yes
BERTScore F1	Form vs Chat	172	0.940	+0.020 negligible	Form > Chat	No
BERTScore F1	Ask vs Chat	59	0.002	−0.664 large	Chat > Ask	Yes
SBERT	Form vs Ask	110	0.099	+0.373 medium	Form > Ask	No
SBERT	Form vs Chat	56	0.002	−0.681 large	Chat > Form	Yes
SBERT	Ask vs Chat	26	< 0.001	−0.852 large	Chat > Ask	Yes

Table D.7: Mann–Whitney U between-groups sensitivity check on the unbalanced full sample of $n = 113$ documents.

Metric	Comparison	p	r	Sig
BERTScore Recall	Form > Ask	< 0.001	0.756 (large)	Yes
BERTScore Recall	Form > Chat	< 0.001	0.512 (large)	Yes
BERTScore F1	Form > Ask	0.004	0.386 (medium)	Yes
BERTScore F1	Chat > Ask	0.011	0.340 (medium)	Yes
SBERT	Chat > Form	< 0.001	0.505 (large)	Yes
SBERT	Chat > Ask	< 0.001	0.453 (large)	Yes
ROUGE-2	Form > Ask	0.006	0.369 (medium)	Yes
ROUGE-2	Chat > Ask	0.011	0.342 (medium)	Yes
METEOR	Form > Ask	0.003	0.407 (medium)	Yes

asures, computed on the survey $\times$ NLP joined sample. Per-mode joined samples: Form $n = 34$, Ask $n = 34$, Chat $n = 38$; pooled $n = 106$.

Table D.8: Spearman rank correlations: primary confidence (5_{ai}) versus document-quality measures, by mode.

Mode	5_{ai} vs BERTScore F1			5_{ai} vs Judge Overall		
	n	ρ	p	n	ρ	p
Form	34	−0.22	0.22	34	+0.07	0.70
Ask	34	−0.23	0.19	34	−0.21	0.24
Chat	38	+0.28	0.09	38	+0.41	0.012
Pooled	106	−0.02	0.88	106	+0.14	0.14

For comparison, the secondary 6-item Perceived AI Support composite produces weaker mode-specific correlations with BERTScore F1 (Form $\rho = -0.22$, Ask $\rho = -0.04$, Chat $\rho = +0.20$) and a small but significant pooled correlation with Judge Overall ($\rho = +0.21$, $p = 0.033$, $n = 106$).

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Declaration

I hereby declare that I have written this thesis independently and have used no sources or aids other than those indicated.

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